

Document 3 - DS - Heart Rate Monitor Application

Application: Heart Rate Monitor for Android and iOS

Document state: DRAFT / uncontrolled / deliberately messy for parser testing

Note: This is synthetic test material only. It is not approved medical-device validation content.

Revision table is mixed below; some rows intentionally repeat or omit values.

Version	Date	Author	Change
0.1	2026-05-18	QA/BA	Initial messy draft
1.0?	TBD	SME?	Imported from old template
0.2	missing	AI Test	duplicate version row

This DS describes the proposed technical design for the mobile app, backend services, local storage, integrations, data model, UI modules, security controls, and monitoring design. Several blocks intentionally contain repeated headings and inconsistent IDs.

DS Section 2 - Cover / uncontrolled draft

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- Old requirement copied from web portal project - review if still valid.
1. Unnumbered note - check sensor disconnect behavior during active monitoring.
 2. Acceptance: expected result to be confirmed by QA; evidence screenshot may be attached later.

ID	Requirement / Test / Step	Area	Risk	Acceptance / Expected Result	Status
DS_2.1	Consent capture and privacy notice	Backend API	Not ranked	Audit/event record shall be generated where applicable.	Draft
DS-002-02	Notification retry and suppression	Cloud Sync	High	Data shall be encrypted in transit and at rest.	Need SME
DS-002-03	Validate sensor data range	Notification Service	Low	Data shall be encrypted in transit and at rest.	In Review
DS-002-04	Export readings as PDF/CSV	Bluetooth Sensor	TBD	System shall complete the action without data loss.	In Review
DS-002-05	Battery and Bluetooth warning	Android	High	User shall receive clear error message on failure.	In Review

DS Section 3 - Abbreviations mixed list

RAG TEST CHUNK

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DS-003-1: App shall process heart rate data and preserve timestamp/timezone context.

- Unnumbered note - check sensor disconnect behavior during active monitoring.

ID	Requirement / Test / Step	Area	Risk	Acceptance / Expected Result	Status
DS-003-01	Support caregiver view	Local DB	TBD	Data shall be encrypted in transit and at rest.	In Review
DS-003-02	Manual heart rate entry when device unavailable	Android/iOS	Not ranked	Audit/event record shall be generated where applicable.	Rejected once
DS-003-03	Validate sensor data range	Android	Medium	Result must be visible in mobile UI within expected time.	Rejected once
DS-003-04	Manual heart rate entry when device unavailable	iOS	Medium	Data shall be encrypted in transit and at rest.	Approved?
DS-003-05	Show trend chart for 7/30/90 days	Local DB	TBD	TBD by clinical SME - intentionally incomplete.	Rejected once

DS Section 4 - Document purpose

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- Acceptance: expected result to be confirmed by QA; evidence screenshot may be attached later.

3. DS-004-2: App shall process heart rate data and preserve timestamp/timezone context.

Unnumbered note - check sensor disconnect behavior during active monitoring.

DS-004-4: App shall process heart rate data and preserve timestamp/timezone context.

ID	Requirement / Test / Step	Area	Risk	Acceptance / Expected Result	Status
DS-004-01	Sync readings to cloud API	Android/iOS	TBD	System shall complete the action without data loss.	In Review
DS-004-02	Consent capture and privacy notice	Local DB	High	User shall receive clear error message on failure.	In Review
DS-004-03	Sync readings to cloud API	Notification Service	Low	System shall complete the action without data loss.	In Review
DS-004-04	Export readings as PDF/CSV	Local DB	TBD	Result must be visible in mobile UI within expected time.	Pending QA
DS-004-05	Capture heart rate from paired device	Android	Low	TBD by clinical SME - intentionally incomplete.	Rejected once

DS Section 5 - Scope and non-scope

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4. Unnumbered note - check sensor disconnect behavior during active monitoring.

For iOS only? For Android only? unclear; parser should capture platform if visible.

- Acceptance: expected result to be confirmed by QA; evidence screenshot may be attached later.

ID	Requirement / Test / Step	Area	Risk	Acceptance / Expected Result	Status
DS-005-01	Handle offline mode	Notification Service	Low	Data shall be encrypted in transit and at rest.	Rejected once
DS_5.2	Handle offline mode	Android/iOS	Medium	System shall complete the action without data loss.	Draft
DS-005-03	Consent capture and privacy notice	Cloud Sync	Not ranked	Result must be visible in mobile UI within expected time.	Need SME
DS-005-04	Sync readings to cloud API	Cloud Sync	Not ranked	Data shall be encrypted in transit and at rest.	Pending QA
DS-005-05	Battery and Bluetooth warning	Cloud Sync	High	System shall complete the action without data loss.	Need SME

DS Section 6 - System overview

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5. Unnumbered note - check sensor disconnect behavior during active monitoring.

Old requirement copied from web portal project - review if still valid.

For iOS only? For Android only? unclear; parser should capture platform if visible.

6. For iOS only? For Android only? unclear; parser should capture platform if visible.

ID	Requirement / Test / Step	Area	Risk	Acceptance / Expected Result	Status
DS-006-01	Manual heart rate entry when device unavailable	Android	Low	TBD by clinical SME - intentionally incomplete.	Approved?
DS-006-02	Show trend chart for 7/30/90 days	Local DB	Not ranked	System shall complete the action without data loss.	In Review
DS-006-03	Consent capture and privacy notice	Local DB	Medium	Audit/event record shall be generated where applicable.	Draft

DS Section 7 - Mobile app overview

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- 7. Unnumbered note - check sensor disconnect behavior during active monitoring.
- 8. DS-007-2: App shall process heart rate data and preserve timestamp/timezone context.

Acceptance: expected result to be confirmed by QA; evidence screenshot may be attached later.

ID	Requirement / Test / Step	Area	Risk	Acceptance / Expected Result	Status
DS_7.1	Validate sensor data range	Android	TBD	Audit/event record shall be generated where applicable.	In Review
DS_7.2	Generate high/low heart-rate alert	Cloud Sync	Low	System shall complete the action without data loss.	Need SME
DS-007-03	Generate high/low heart-rate alert	iOS	TBD	Result must be visible in mobile UI within expected time.	In Review
DS-007-04	Sync readings to cloud API	Bluetooth Sensor	Low	System shall complete the action without data loss.	Rejected once

DS Section 8 - User roles

duplicate section

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Unnumbered note - check sensor disconnect behavior during active monitoring.

DS-008-2: App shall process heart rate data and preserve timestamp/timezone context.

DS-008-3: App shall process heart rate data and preserve timestamp/timezone context.

9. Unnumbered note - check sensor disconnect behavior during active monitoring.

ID	Requirement / Test / Step	Area	Risk	Acceptance / Expected Result	Status
DS-008-01	Display current BPM with timestamp	Local DB	High	Audit/event record shall be generated where applicable.	Rejected once
DS-008-02	Battery and Bluetooth warning	Android	High	Result must be visible in mobile UI within expected time.	Pending QA
DS-008-03	Generate high/low heart-rate alert	iOS	Not ranked	TBD by clinical SME - intentionally incomplete.	Rejected once

DS Section 9 - Regulatory notes

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DS-009-1: App shall process heart rate data and preserve timestamp/timezone context.

10. For iOS only? For Android only? unclear; parser should capture platform if visible.

ID	Requirement / Test / Step	Area	Risk	Acceptance / Expected Result	Status
DS-009-01	Generate high/low heart-rate alert	Backend API	Low	Result must be visible in mobile UI within expected time.	Approved?
DS-009-02	Export readings as PDF/CSV	Cloud Sync	Low	Audit/event record shall be generated where applicable.	Pending QA
DS-009-03	Biometric login option	Android/iOS	Medium	TBD by clinical SME - intentionally incomplete.	In Review
DS_9.4	Validate sensor data range	Notification Service	Low	Audit/event record shall be generated where applicable.	Pending QA

MIXED TRACEABILITY LINE: URS-00X -> FRS?? -> DS component maybe -> IQ/OQ/PQ evidence TBD. This line is intentionally noisy.

DS Section 10 - Data privacy notes

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11. Acceptance: expected result to be confirmed by QA; evidence screenshot may be attached later.

- DS-010-2: App shall process heart rate data and preserve timestamp/timezone context.

Old requirement copied from web portal project - review if still valid.

ID	Requirement / Test / Step	Area	Risk	Acceptance / Expected Result	Status
DS-010-01	Export readings as PDF/CSV	Android/iOS	Not ranked	TBD by clinical SME - intentionally incomplete.	Rejected once
DS-010-02	Capture heart rate from paired device	Bluetooth Sensor	TBD	User shall receive clear error message on failure.	Need SME
DS-010-03	Consent capture and privacy notice	Notification Service	Not ranked	Result must be visible in mobile UI within expected time.	Rejected once
DS-010-04	Sync readings to cloud API	Notification Service	Medium	Result must be visible in mobile UI within expected time.	In Review
DS-010-05	Manual heart rate entry when device unavailable	Android	Medium	System shall complete the action without data loss.	In Review

DS Section 11 - High level assumptions

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For iOS only? For Android only? unclear; parser should capture platform if visible.

- 12. Unnumbered note - check sensor disconnect behavior during active monitoring.
- 13. Old requirement copied from web portal project - review if still valid.

ID	Requirement / Test / Step	Area	Risk	Acceptance / Expected Result	Status
DS-011-01	Manual heart rate entry when device unavailable	Android/iOS	Medium	TBD by clinical SME - intentionally incomplete.	In Review
DS-011-02	Sync readings to cloud API	Notification Service	TBD	Data shall be encrypted in transit and at rest.	Rejected once
DS-011-03	Capture heart rate from paired device	Backend API	High	TBD by clinical SME - intentionally incomplete.	Approved?

DS Section 12 - Cover / uncontrolled draft
old text retained

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- Unnumbered note - check sensor disconnect behavior during active monitoring.

Old requirement copied from web portal project - review if still valid.

ID	Requirement / Test / Step	Area	Risk	Acceptance / Expected Result	Status
DS-012-01	Generate high/low heart-rate alert	Local DB	Not ranked	Audit/event record shall be generated where applicable.	Approved?
DS-012-02	Biometric login option	Backend API	Low	Result must be visible in mobile UI within expected time.	Approved?
DS-012-03	Show trend chart for 7/30/90 days	Local DB	Not ranked	User shall receive clear error message on failure.	Draft
DS-012-04	Consent capture and privacy notice	Android/iOS	Medium	User shall receive clear error message on failure.	Need SME
DS-012-05	Notification retry and suppression	Local DB	Not ranked	System shall complete the action without data loss.	Need SME

DS Section 13 - Abbreviations mixed list

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Old requirement copied from web portal project - review if still valid.

Unnumbered note - check sensor disconnect behavior during active monitoring.

- Old requirement copied from web portal project - review if still valid.

ID	Requirement / Test / Step	Area	Risk	Acceptance / Expected Result	Status
DS-013-01	Sync readings to cloud API	Local DB	Not ranked	Audit/event record shall be generated where applicable.	In Review
DS-013-02	Export readings as PDF/CSV	iOS	TBD	Result must be visible in mobile UI within expected time.	In Review
DS-013-03	Audit user actions	Bluetooth Sensor	High	Result must be visible in mobile UI within expected time.	Approved?
DS-013-04	Generate high/low heart-rate alert	Android/iOS	Low	System shall complete the action without data loss.	Draft

Random inserted note: duplicate title appears again below.

DS Section 13 - Abbreviations mixed list

DS Section 14 - Document purpose

duplicate section

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DS-014-1: App shall process heart rate data and preserve timestamp/timezone context.

- Old requirement copied from web portal project - review if still valid.
- Unnumbered note - check sensor disconnect behavior during active monitoring.

ID	Requirement / Test / Step	Area	Risk	Acceptance / Expected Result	Status
DS_14.1	Manual heart rate entry when device unavailable	Android	TBD	TBD by clinical SME - intentionally incomplete.	Need SME
DS-014-02	Handle offline mode	Android/iOS	TBD	User shall receive clear error message on failure.	Draft
DS_14.3	Display current BPM with timestamp	Bluetooth Sensor	Low	TBD by clinical SME - intentionally incomplete.	Need SME
DS_14.4	Export readings as PDF/CSV	iOS	Low	System shall complete the action without data loss.	Pending QA

DS Section 15 - Test condition table

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- For iOS only? For Android only? unclear; parser should capture platform if visible.
- DS-015-2: App shall process heart rate data and preserve timestamp/timezone context.

For iOS only? For Android only? unclear; parser should capture platform if visible.

ID	Requirement / Test / Step	Area	Risk	Acceptance / Expected Result	Status
DS-015-01	Sync readings to cloud API	Backend API	Low	System shall complete the action without data loss.	Draft
DS_15.2	Biometric login option	Backend API	Medium	Result must be visible in mobile UI within expected time.	In Review
DS-015-03	Notification retry and suppression	iOS	High	TBD by clinical SME - intentionally incomplete.	In Review

DS Section 16 - API behavior

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Unnumbered note - check sensor disconnect behavior during active monitoring.

- 14. Old requirement copied from web portal project - review if still valid.
- 15. Unnumbered note - check sensor disconnect behavior during active monitoring.

ID	Requirement / Test / Step	Area	Risk	Acceptance / Expected Result	Status
DS-016-01	Consent capture and privacy notice	Cloud Sync	TBD	Audit/event record shall be generated where applicable.	Rejected once
DS-016-02	Sync readings to cloud API	Cloud Sync	TBD	Result must be visible in mobile UI within expected time.	Pending QA
DS-016-03	Show trend chart for 7/30/90 days	Notification Service	High	Data shall be encrypted in transit and at rest.	Pending QA

DS Section 17 - Test condition table

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- Old requirement copied from web portal project - review if still valid.
- For iOS only? For Android only? unclear; parser should capture platform if visible.
- For iOS only? For Android only? unclear; parser should capture platform if visible.

ID	Requirement / Test / Step	Area	Risk	Acceptance / Expected Result	Status
DS-017-01	Battery and Bluetooth warning	Cloud Sync	Medium	TBD by clinical SME - intentionally incomplete.	In Review
DS-017-02	Battery and Bluetooth warning	Bluetooth Sensor	Medium	Audit/event record shall be generated where applicable.	Pending QA
DS-017-03	Manual heart rate entry when device unavailable	Cloud Sync	TBD	TBD by clinical SME - intentionally incomplete.	Rejected once
DS-017-04	Validate sensor data range	Local DB	Medium	User shall receive clear error message on failure.	Pending QA

DS Section 18 - Mobile platform variation

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For iOS only? For Android only? unclear; parser should capture platform if visible.

Acceptance: expected result to be confirmed by QA; evidence screenshot may be attached later.

- For iOS only? For Android only? unclear; parser should capture platform if visible.

ID	Requirement / Test / Step	Area	Risk	Acceptance / Expected Result	Status
DS_18.1	Handle offline mode	Android	Low	TBD by clinical SME - intentionally incomplete.	Draft
DS-018-02	Export readings as PDF/CSV	iOS	TBD	TBD by clinical SME - intentionally incomplete.	In Review
DS_18.3	Capture heart rate from paired device	Android	Not ranked	Audit/event record shall be generated where applicable.	Rejected once
DS-018-04	Biometric login option	Android/iOS	Not ranked	Result must be visible in mobile UI within expected time.	Rejected once

MIXED TRACEABILITY LINE: URS-00X -> FRS?? -> DS component maybe -> IQ/OQ/PQ evidence TBD. This line is intentionally noisy.

DS Section 19 - Security and privacy

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16. For iOS only? For Android only? unclear; parser should capture platform if visible.

Acceptance: expected result to be confirmed by QA; evidence screenshot may be attached later.

17. For iOS only? For Android only? unclear; parser should capture platform if visible.

For iOS only? For Android only? unclear; parser should capture platform if visible.

ID	Requirement / Test / Step	Area	Risk	Acceptance / Expected Result	Status
DS-019-01	Manual heart rate entry when device unavailable	Local DB	High	Result must be visible in mobile UI within expected time.	Rejected once
DS-019-02	Consent capture and privacy notice	Cloud Sync	TBD	User shall receive clear error message on failure.	Draft
DS-019-03	Notification retry and suppression	Android/iOS	High	System shall complete the action without data loss.	Draft
DS-019-04	Handle offline mode	Cloud Sync	TBD	Audit/event record shall be generated where applicable.	Pending QA
DS-019-05	Store readings securely on device	Bluetooth Sensor	Not ranked	Audit/event record shall be generated where applicable.	In Review

DS Section 20 - Open questions

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Acceptance: expected result to be confirmed by QA; evidence screenshot may be attached later.

Unnumbered note - check sensor disconnect behavior during active monitoring.

For iOS only? For Android only? unclear; parser should capture platform if visible.

18. For iOS only? For Android only? unclear; parser should capture platform if visible.

ID	Requirement / Test / Step	Area	Risk	Acceptance / Expected Result	Status
DS-020-01	Biometric login option	Android/iOS	High	Data shall be encrypted in transit and at rest.	Rejected once
DS-020-02	Support caregiver view	Local DB	Medium	User shall receive clear error message on failure.	Draft
DS-020-03	Store readings securely on device	Cloud Sync	Not ranked	Audit/event record shall be generated where applicable.	Pending QA
DS-020-04	Biometric login option	Android	TBD	System shall complete the action without data loss.	Need SME
DS-020-05	Store readings securely on device	Notification Service	Low	User shall receive clear error message on failure.	Draft

DS Section 21 - Open questions

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- 19. Old requirement copied from web portal project - review if still valid.
- 20. For iOS only? For Android only? unclear; parser should capture platform if visible.

ID	Requirement / Test / Step	Area	Risk	Acceptance / Expected Result	Status
DS-021-01	Consent capture and privacy notice	Bluetooth Sensor	TBD	Result must be visible in mobile UI within expected time.	Draft
DS-021-02	Biometric login option	Cloud Sync	Medium	Audit/event record shall be generated where applicable.	In Review
DS-021-03	Store readings securely on device	Local DB	Low	Data shall be encrypted in transit and at rest.	In Review
DS-021-04	Support caregiver view	Cloud Sync	TBD	System shall complete the action without data loss.	Pending QA

DS Section 22 - Data handling notes

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21. For iOS only? For Android only? unclear; parser should capture platform if visible.
- Acceptance: expected result to be confirmed by QA; evidence screenshot may be attached later.

ID	Requirement / Test / Step	Area	Risk	Acceptance / Expected Result	Status
DS-022-01	Store readings securely on device	Notification Service	Medium	System shall complete the action without data loss.	Draft
DS_22.2	Handle offline mode	Cloud Sync	Medium	Audit/event record shall be generated where applicable.	Approved?
DS-022-03	Manual heart rate entry when device unavailable	Backend API	Medium	Data shall be encrypted in transit and at rest.	In Review

DS Section 23 - Open questions

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- Acceptance: expected result to be confirmed by QA; evidence screenshot may be attached later.
22. Acceptance: expected result to be confirmed by QA; evidence screenshot may be attached later.
23. For iOS only? For Android only? unclear; parser should capture platform if visible.
- DS-023-4: App shall process heart rate data and preserve timestamp/timezone context.

ID	Requirement / Test / Step	Area	Risk	Acceptance / Expected Result	Status
DS-023-01	Store readings securely on device	Android/iOS	Low	User shall receive clear error message on failure.	Rejected once
DS-023-02	Export readings as PDF/CSV	Local DB	High	Audit/event record shall be generated where applicable.	Draft
DS-023-03	Battery and Bluetooth warning	Backend API	Not ranked	TBD by clinical SME - intentionally incomplete.	In Review
DS_23.4	Handle offline mode	iOS	Not ranked	Data shall be encrypted in transit and at rest.	Pending QA

DS Section 24 - Mobile platform variation

RAG TEST CHUNK

This page belongs to Design Specification for the Heart Rate Monitor mobile application. The application supports Android and iOS users, paired heart-rate sensors, manual BPM entry, trend review, alerts, data export, and controlled access. This block intentionally includes uneven numbering, repeated labels, partial owner fields, and mixed wording so an AI traceability/RAG engine can be tested against non-perfect documents. Page marker: DS-024. Owner: Product. Important terms: BPM, sensor pairing, low alert, high alert, caregiver, audit trail, local cache, encrypted sync, consent, export, notification.

For iOS only? For Android only? unclear; parser should capture platform if visible.

- Acceptance: expected result to be confirmed by QA; evidence screenshot may be attached later.

ID	Requirement / Test / Step	Area	Risk	Acceptance / Expected Result	Status
DS-024-01	Display current BPM with timestamp	Bluetooth Sensor	Low	User shall receive clear error message on failure.	In Review
DS-024-02	Support caregiver view	iOS	Medium	Data shall be encrypted in transit and at rest.	Need SME
DS-024-03	Capture heart rate from paired device	Backend API	High	User shall receive clear error message on failure.	In Review
DS_24.4	Manual heart rate entry when device unavailable	Cloud Sync	Medium	User shall receive clear error message on failure.	Need SME

DS Section 25 - Bluetooth edge cases

missing owner

This page belongs to Design Specification for the Heart Rate Monitor mobile application. The application supports Android and iOS users, paired heart-rate sensors, manual BPM entry, trend review, alerts, data export, and controlled access. This block intentionally includes uneven numbering, repeated labels, partial owner fields, and mixed wording so an AI traceability/RAG engine can be tested against non-perfect documents. Page marker: DS-025. Owner: Validation. Important terms: BPM, sensor pairing, low alert, high alert, caregiver, audit trail, local cache, encrypted sync, consent, export, notification.

DS-025-1: App shall process heart rate data and preserve timestamp/timezone context.

Unnumbered note - check sensor disconnect behavior during active monitoring.

24. Old requirement copied from web portal project - review if still valid.

Acceptance: expected result to be confirmed by QA; evidence screenshot may be attached later.

ID	Requirement / Test / Step	Area	Risk	Acceptance / Expected Result	Status
DS-025-01	Capture heart rate from paired device	Android/iOS	Low	User shall receive clear error message on failure.	In Review
DS-025-02	Handle offline mode	Local DB	Low	User shall receive clear error message on failure.	In Review
DS-025-03	Generate high/low heart-rate alert	Local DB	Not ranked	Data shall be encrypted in transit and at rest.	Approved?

DS Section 26 - Exception handling

This page belongs to Design Specification for the Heart Rate Monitor mobile application. The application supports Android and iOS users, paired heart-rate sensors, manual BPM entry, trend review, alerts, data export, and controlled access. This block intentionally includes uneven numbering, repeated labels, partial owner fields, and mixed wording so an AI traceability/RAG engine can be tested against non-perfect documents. Page marker: DS-026. Owner: Product. Important terms: BPM, sensor pairing, low alert, high alert, caregiver, audit trail, local cache, encrypted sync, consent, export, notification.

Old requirement copied from web portal project - review if still valid.

25. Old requirement copied from web portal project - review if still valid.

ID	Requirement / Test / Step	Area	Risk	Acceptance / Expected Result	Status
DS_26.1	Audit user actions	Android/iOS	TBD	User shall receive clear error message on failure.	Draft
DS-026-02	Generate high/low heart-rate alert	iOS	Not ranked	User shall receive clear error message on failure.	Need SME
DS-026-03	Export readings as PDF/CSV	Local DB	High	Audit/event record shall be generated where applicable.	Need SME

Random inserted note: duplicate title appears again below.

DS Section 26 - Exception handling

DS Section 27 - API behavior

This page belongs to Design Specification for the Heart Rate Monitor mobile application. The application supports Android and iOS users, paired heart-rate sensors, manual BPM entry, trend review, alerts, data export, and controlled access. This block intentionally includes uneven numbering, repeated labels, partial owner fields, and mixed wording so an AI traceability/RAG engine can be tested against non-perfect documents. Page marker: DS-027. Owner: TBD. Important terms: BPM, sensor pairing, low alert, high alert, caregiver, audit trail, local cache, encrypted sync, consent, export, notification.

Old requirement copied from web portal project - review if still valid.

DS-027-2: App shall process heart rate data and preserve timestamp/timezone context.

26. DS-027-3: App shall process heart rate data and preserve timestamp/timezone context.

ID	Requirement / Test / Step	Area	Risk	Acceptance / Expected Result	Status
DS-027-01	Show trend chart for 7/30/90 days	Backend API	TBD	User shall receive clear error message on failure.	Approved?
DS_27.2	Consent capture and privacy notice	Local DB	Medium	Data shall be encrypted in transit and at rest.	Rejected once
DS-027-03	Sync readings to cloud API	Android/iOS	TBD	User shall receive clear error message on failure.	Pending QA

MIXED TRACEABILITY LINE: URS-00X -> FRS?? -> DS component maybe -> IQ/OQ/PQ evidence TBD. This line is intentionally noisy.

DS Section 28 - UI behavior

This page belongs to Design Specification for the Heart Rate Monitor mobile application. The application supports Android and iOS users, paired heart-rate sensors, manual BPM entry, trend review, alerts, data export, and controlled access. This block intentionally includes uneven numbering, repeated labels, partial owner fields, and mixed wording so an AI traceability/RAG engine can be tested against non-perfect documents. Page marker: DS-028. Owner: Mobile Lead. Important terms: BPM, sensor pairing, low alert, high alert, caregiver, audit trail, local cache, encrypted sync, consent, export, notification.

- 27. Acceptance: expected result to be confirmed by QA; evidence screenshot may be attached later.
- 28. Acceptance: expected result to be confirmed by QA; evidence screenshot may be attached later.

For iOS only? For Android only? unclear; parser should capture platform if visible.

ID	Requirement / Test / Step	Area	Risk	Acceptance / Expected Result	Status
DS-028-01	Notification retry and suppression	Local DB	Low	User shall receive clear error message on failure.	In Review
DS_28.2	Generate high/low heart-rate alert	Cloud Sync	Medium	Audit/event record shall be generated where applicable.	Approved?
DS-028-03	Audit user actions	Notification Service	High	Audit/event record shall be generated where applicable.	Need SME
DS-028-04	Validate sensor data range	Android/iOS	High	Result must be visible in mobile UI within expected time.	Pending QA
DS-028-05	Battery and Bluetooth warning	Cloud Sync	Medium	User shall receive clear error message on failure.	Draft

DS Section 29 - Requirement details

OCR noisy line

This page belongs to Design Specification for the Heart Rate Monitor mobile application. The application supports Android and iOS users, paired heart-rate sensors, manual BPM entry, trend review, alerts, data export, and controlled access. This block intentionally includes uneven numbering, repeated labels, partial owner fields, and mixed wording so an AI traceability/RAG engine can be tested against non-perfect documents. Page marker: DS-029. Owner: TBD. Important terms: BPM, sensor pairing, low alert, high alert, caregiver, audit trail, local cache, encrypted sync, consent, export, notification.

29. Unnumbered note - check sensor disconnect behavior during active monitoring.

For iOS only? For Android only? unclear; parser should capture platform if visible.

ID	Requirement / Test / Step	Area	Risk	Acceptance / Expected Result	Status
DS_29.1	Handle offline mode	Local DB	Medium	Data shall be encrypted in transit and at rest.	Approved?
DS-029-02	Battery and Bluetooth warning	Notification Service	Low	TBD by clinical SME - intentionally incomplete.	Pending QA
DS-029-03	Audit user actions	Android/iOS	High	User shall receive clear error message on failure.	Pending QA
DS-029-04	Show trend chart for 7/30/90 days	Bluetooth Sensor	High	Data shall be encrypted in transit and at rest.	Approved?
DS-029-05	Display current BPM with timestamp	Android	TBD	Audit/event record shall be generated where applicable.	Draft

DS Section 30 - UI behavior

This page belongs to Design Specification for the Heart Rate Monitor mobile application. The application supports Android and iOS users, paired heart-rate sensors, manual BPM entry, trend review, alerts, data export, and controlled access. This block intentionally includes uneven numbering, repeated labels, partial owner fields, and mixed wording so an AI traceability/RAG engine can be tested against non-perfect documents. Page marker: DS-030. Owner: Clinical SME. Important terms: BPM, sensor pairing, low alert, high alert, caregiver, audit trail, local cache, encrypted sync, consent, export, notification.

- For iOS only? For Android only? unclear; parser should capture platform if visible.
- Unnumbered note - check sensor disconnect behavior during active monitoring.

ID	Requirement / Test / Step	Area	Risk	Acceptance / Expected Result	Status
DS-030-01	Store readings securely on device	Android/iOS	TBD	TBD by clinical SME - intentionally incomplete.	Need SME
DS_30.2	Display current BPM with timestamp	Cloud Sync	Not ranked	Result must be visible in mobile UI within expected time.	Pending QA
DS-030-03	Validate sensor data range	Android/iOS	High	Result must be visible in mobile UI within expected time.	Need SME

DS Section 31 - Data handling notes

This page belongs to Design Specification for the Heart Rate Monitor mobile application. The application supports Android and iOS users, paired heart-rate sensors, manual BPM entry, trend review, alerts, data export, and controlled access. This block intentionally includes uneven numbering, repeated labels, partial owner fields, and mixed wording so an AI traceability/RAG engine can be tested against non-perfect documents. Page marker: DS-031. Owner: QA. Important terms: BPM, sensor pairing, low alert, high alert, caregiver, audit trail, local cache, encrypted sync, consent, export, notification.

- Acceptance: expected result to be confirmed by QA; evidence screenshot may be attached later.

30. Acceptance: expected result to be confirmed by QA; evidence screenshot may be attached later.

Acceptance: expected result to be confirmed by QA; evidence screenshot may be attached later.

- Old requirement copied from web portal project - review if still valid.

ID	Requirement / Test / Step	Area	Risk	Acceptance / Expected Result	Status
DS-031-01	Audit user actions	Notification Service	Not ranked	Audit/event record shall be generated where applicable.	Pending QA
DS-031-02	Support caregiver view	Android/iOS	Medium	Audit/event record shall be generated where applicable.	Approved?
DS-031-03	Consent capture and privacy notice	Cloud Sync	High	Result must be visible in mobile UI within expected time.	In Review
DS-031-04	Store readings securely on device	iOS	Low	Data shall be encrypted in transit and at rest.	Rejected once
DS-031-05	Consent capture and privacy notice	Cloud Sync	High	User shall receive clear error message on failure.	Rejected once

DS Section 32 - Exception handling

This page belongs to Design Specification for the Heart Rate Monitor mobile application. The application supports Android and iOS users, paired heart-rate sensors, manual BPM entry, trend review, alerts, data export, and controlled access. This block intentionally includes uneven numbering, repeated labels, partial owner fields, and mixed wording so an AI traceability/RAG engine can be tested against non-perfect documents. Page marker: DS-032. Owner: QA. Important terms: BPM, sensor pairing, low alert, high alert, caregiver, audit trail, local cache, encrypted sync, consent, export, notification.

- Old requirement copied from web portal project - review if still valid.

DS-032-2: App shall process heart rate data and preserve timestamp/timezone context.

Unnumbered note - check sensor disconnect behavior during active monitoring.

ID	Requirement / Test / Step	Area	Risk	Acceptance / Expected Result	Status
DS-032-01	Manual heart rate entry when device unavailable	Backend API	Low	TBD by clinical SME - intentionally incomplete.	Rejected once
DS-032-02	Manual heart rate entry when device unavailable	Cloud Sync	High	Result must be visible in mobile UI within expected time.	Need SME
DS-032-03	Validate sensor data range	Local DB	Low	User shall receive clear error message on failure.	Need SME

DS Section 33 - Bluetooth edge cases

This page belongs to Design Specification for the Heart Rate Monitor mobile application. The application supports Android and iOS users, paired heart-rate sensors, manual BPM entry, trend review, alerts, data export, and controlled access. This block intentionally includes uneven numbering, repeated labels, partial owner fields, and mixed wording so an AI traceability/RAG engine can be tested against non-perfect documents. Page marker: DS-033. Owner: Product. Important terms: BPM, sensor pairing, low alert, high alert, caregiver, audit trail, local cache, encrypted sync, consent, export, notification.

- Unnumbered note - check sensor disconnect behavior during active monitoring.

DS-033-2: App shall process heart rate data and preserve timestamp/timezone context.

For iOS only? For Android only? unclear; parser should capture platform if visible.

ID	Requirement / Test / Step	Area	Risk	Acceptance / Expected Result	Status
DS_33.1	Manual heart rate entry when device unavailable	iOS	Not ranked	TBD by clinical SME - intentionally incomplete.	Rejected once
DS-033-02	Biometric login option	Backend API	High	Data shall be encrypted in transit and at rest.	Draft
DS-033-03	Display current BPM with timestamp	Android	Low	TBD by clinical SME - intentionally incomplete.	Draft
DS_33.4	Sync readings to cloud API	Backend API	Medium	User shall receive clear error message on failure.	Pending QA
DS_33.5	Generate high/low heart-rate alert	Cloud Sync	Medium	Result must be visible in mobile UI within expected time.	Approved?

DS Section 34 - API behavior

This page belongs to Design Specification for the Heart Rate Monitor mobile application. The application supports Android and iOS users, paired heart-rate sensors, manual BPM entry, trend review, alerts, data export, and controlled access. This block intentionally includes uneven numbering, repeated labels, partial owner fields, and mixed wording so an AI traceability/RAG engine can be tested against non-perfect documents. Page marker: DS-034. Owner: QA. Important terms: BPM, sensor pairing, low alert, high alert, caregiver, audit trail, local cache, encrypted sync, consent, export, notification.

31. Unnumbered note - check sensor disconnect behavior during active monitoring.
- DS-034-2: App shall process heart rate data and preserve timestamp/timezone context.

ID	Requirement / Test / Step	Area	Risk	Acceptance / Expected Result	Status
DS-034-01	Notification retry and suppression	iOS	High	TBD by clinical SME - intentionally incomplete.	Need SME
DS-034-02	Store readings securely on device	Android	TBD	System shall complete the action without data loss.	Draft
DS-034-03	Capture heart rate from paired device	Cloud Sync	TBD	Result must be visible in mobile UI within expected time.	Pending QA
DS-034-04	Handle offline mode	Android	Medium	System shall complete the action without data loss.	Approved?

DS Section 35 - Security and privacy

This page belongs to Design Specification for the Heart Rate Monitor mobile application. The application supports Android and iOS users, paired heart-rate sensors, manual BPM entry, trend review, alerts, data export, and controlled access. This block intentionally includes uneven numbering, repeated labels, partial owner fields, and mixed wording so an AI traceability/RAG engine can be tested against non-perfect documents. Page marker: DS-035. Owner: Clinical SME. Important terms: BPM, sensor pairing, low alert, high alert, caregiver, audit trail, local cache, encrypted sync, consent, export, notification.

Old requirement copied from web portal project - review if still valid.

- 32. For iOS only? For Android only? unclear; parser should capture platform if visible.
- 33. Acceptance: expected result to be confirmed by QA; evidence screenshot may be attached later.

Unnumbered note - check sensor disconnect behavior during active monitoring.

ID	Requirement / Test / Step	Area	Risk	Acceptance / Expected Result	Status
DS-035-01	Support caregiver view	Bluetooth Sensor	Low	Data shall be encrypted in transit and at rest.	Rejected once
DS_35.2	Store readings securely on device	Android	Medium	User shall receive clear error message on failure.	Pending QA
DS-035-03	Validate sensor data range	Backend API	Low	Result must be visible in mobile UI within expected time.	Rejected once
DS_35.4	Export readings as PDF/CSV	Local DB	TBD	TBD by clinical SME - intentionally incomplete.	Approved?

DS Section 36 - Open questions

This page belongs to Design Specification for the Heart Rate Monitor mobile application. The application supports Android and iOS users, paired heart-rate sensors, manual BPM entry, trend review, alerts, data export, and controlled access. This block intentionally includes uneven numbering, repeated labels, partial owner fields, and mixed wording so an AI traceability/RAG engine can be tested against non-perfect documents. Page marker: DS-036. Owner: Validation. Important terms: BPM, sensor pairing, low alert, high alert, caregiver, audit trail, local cache, encrypted sync, consent, export, notification.

DS-036-1: App shall process heart rate data and preserve timestamp/timezone context.

34. Acceptance: expected result to be confirmed by QA; evidence screenshot may be attached later.

- Acceptance: expected result to be confirmed by QA; evidence screenshot may be attached later.
- Acceptance: expected result to be confirmed by QA; evidence screenshot may be attached later.

ID	Requirement / Test / Step	Area	Risk	Acceptance / Expected Result	Status
DS-036-01	Notification retry and suppression	Notification Service	Not ranked	TBD by clinical SME - intentionally incomplete.	Draft
DS-036-02	Sync readings to cloud API	Android/iOS	Not ranked	TBD by clinical SME - intentionally incomplete.	Rejected once
DS-036-03	Audit user actions	Android	Low	System shall complete the action without data loss.	Rejected once
DS-036-04	Capture heart rate from paired device	Android/iOS	Not ranked	TBD by clinical SME - intentionally incomplete.	Rejected once

MIXED TRACEABILITY LINE: URS-00X -> FRS?? -> DS component maybe -> IQ/OQ/PQ evidence TBD. This line is intentionally noisy.

DS Section 37 - Repeated requirement block

This page belongs to Design Specification for the Heart Rate Monitor mobile application. The application supports Android and iOS users, paired heart-rate sensors, manual BPM entry, trend review, alerts, data export, and controlled access. This block intentionally includes uneven numbering, repeated labels, partial owner fields, and mixed wording so an AI traceability/RAG engine can be tested against non-perfect documents. Page marker: DS-037. Owner: Product. Important terms: BPM, sensor pairing, low alert, high alert, caregiver, audit trail, local cache, encrypted sync, consent, export, notification.

35. Unnumbered note - check sensor disconnect behavior during active monitoring.

- Old requirement copied from web portal project - review if still valid.

For iOS only? For Android only? unclear; parser should capture platform if visible.

ID	Requirement / Test / Step	Area	Risk	Acceptance / Expected Result	Status
DS-037-01	Generate high/low heart-rate alert	Cloud Sync	Medium	Result must be visible in mobile UI within expected time.	Approved?
DS-037-02	Audit user actions	Backend API	Medium	System shall complete the action without data loss.	Need SME
DS-037-03	Consent capture and privacy notice	Notification Service	Medium	Result must be visible in mobile UI within expected time.	Rejected once
DS-037-04	Consent capture and privacy notice	Cloud Sync	TBD	User shall receive clear error message on failure.	Rejected once
DS-037-05	Validate sensor data range	Android	High	TBD by clinical SME - intentionally incomplete.	In Review

DS Section 38 - Requirement details

duplicate section

This page belongs to Design Specification for the Heart Rate Monitor mobile application. The application supports Android and iOS users, paired heart-rate sensors, manual BPM entry, trend review, alerts, data export, and controlled access. This block intentionally includes uneven numbering, repeated labels, partial owner fields, and mixed wording so an AI traceability/RAG engine can be tested against non-perfect documents. Page marker: DS-038. Owner: Clinical SME. Important terms: BPM, sensor pairing, low alert, high alert, caregiver, audit trail, local cache, encrypted sync, consent, export, notification.

For iOS only? For Android only? unclear; parser should capture platform if visible.

- For iOS only? For Android only? unclear; parser should capture platform if visible.

ID	Requirement / Test / Step	Area	Risk	Acceptance / Expected Result	Status
DS-038-01	Biometric login option	Android/iOS	High	TBD by clinical SME - intentionally incomplete.	Approved?
DS-038-02	Handle offline mode	Backend API	Not ranked	Result must be visible in mobile UI within expected time.	Need SME
DS-038-03	Manual heart rate entry when device unavailable	Notification Service	High	Result must be visible in mobile UI within expected time.	In Review
DS_38.4	Store readings securely on device	Android/iOS	Not ranked	System shall complete the action without data loss.	Pending QA

DS Section 39 - Mobile platform variation

TBD

This page belongs to Design Specification for the Heart Rate Monitor mobile application. The application supports Android and iOS users, paired heart-rate sensors, manual BPM entry, trend review, alerts, data export, and controlled access. This block intentionally includes uneven numbering, repeated labels, partial owner fields, and mixed wording so an AI traceability/RAG engine can be tested against non-perfect documents. Page marker: DS-039. Owner: Clinical SME. Important terms: BPM, sensor pairing, low alert, high alert, caregiver, audit trail, local cache, encrypted sync, consent, export, notification.

Unnumbered note - check sensor disconnect behavior during active monitoring.

36. Old requirement copied from web portal project - review if still valid.

ID	Requirement / Test / Step	Area	Risk	Acceptance / Expected Result	Status
DS-039-01	Notification retry and suppression	iOS	High	Result must be visible in mobile UI within expected time.	Need SME
DS_39.2	Store readings securely on device	Backend API	Low	Audit/event record shall be generated where applicable.	Draft
DS-039-03	Manual heart rate entry when device unavailable	Notification Service	Medium	TBD by clinical SME - intentionally incomplete.	Approved?

Random inserted note: duplicate title appears again below.

DS Section 39 - Mobile platform variation

DS Section 40 - Risk and control mapping

REVIEW-LATER

This page belongs to Design Specification for the Heart Rate Monitor mobile application. The application supports Android and iOS users, paired heart-rate sensors, manual BPM entry, trend review, alerts, data export, and controlled access. This block intentionally includes uneven numbering, repeated labels, partial owner fields, and mixed wording so an AI traceability/RAG engine can be tested against non-perfect documents. Page marker: DS-040. Owner: Product. Important terms: BPM, sensor pairing, low alert, high alert, caregiver, audit trail, local cache, encrypted sync, consent, export, notification.

- For iOS only? For Android only? unclear; parser should capture platform if visible.

Old requirement copied from web portal project - review if still valid.

Unnumbered note - check sensor disconnect behavior during active monitoring.

ID	Requirement / Test / Step	Area	Risk	Acceptance / Expected Result	Status
DS-040-01	Generate high/low heart-rate alert	Android/iOS	Medium	TBD by clinical SME - intentionally incomplete.	Draft
DS-040-02	Manual heart rate entry when device unavailable	Android/iOS	Low	System shall complete the action without data loss.	Pending QA
DS-040-03	Biometric login option	Android/iOS	Medium	TBD by clinical SME - intentionally incomplete.	Draft
DS-040-04	Audit user actions	Local DB	TBD	Result must be visible in mobile UI within expected time.	Pending QA

DS Section 41 - Risk and control mapping

This page belongs to Design Specification for the Heart Rate Monitor mobile application. The application supports Android and iOS users, paired heart-rate sensors, manual BPM entry, trend review, alerts, data export, and controlled access. This block intentionally includes uneven numbering, repeated labels, partial owner fields, and mixed wording so an AI traceability/RAG engine can be tested against non-perfect documents. Page marker: DS-041. Owner: Product. Important terms: BPM, sensor pairing, low alert, high alert, caregiver, audit trail, local cache, encrypted sync, consent, export, notification.

37. For iOS only? For Android only? unclear; parser should capture platform if visible.

Old requirement copied from web portal project - review if still valid.

DS-041-3: App shall process heart rate data and preserve timestamp/timezone context.

- DS-041-4: App shall process heart rate data and preserve timestamp/timezone context.

ID	Requirement / Test / Step	Area	Risk	Acceptance / Expected Result	Status
DS-041-01	Capture heart rate from paired device	Android	TBD	TBD by clinical SME - intentionally incomplete.	Need SME
DS-041-02	Export readings as PDF/CSV	Android/iOS	High	Result must be visible in mobile UI within expected time.	In Review
DS_41.3	Consent capture and privacy notice	Cloud Sync	Not ranked	Data shall be encrypted in transit and at rest.	Draft
DS_41.4	Show trend chart for 7/30/90 days	Bluetooth Sensor	TBD	Data shall be encrypted in transit and at rest.	Rejected once
DS-041-05	Consent capture and privacy notice	iOS	Not ranked	User shall receive clear error message on failure.	Pending QA

DS Section 42 - Evidence placeholder

This page belongs to Design Specification for the Heart Rate Monitor mobile application. The application supports Android and iOS users, paired heart-rate sensors, manual BPM entry, trend review, alerts, data export, and controlled access. This block intentionally includes uneven numbering, repeated labels, partial owner fields, and mixed wording so an AI traceability/RAG engine can be tested against non-perfect documents. Page marker: DS-042. Owner: Product. Important terms: BPM, sensor pairing, low alert, high alert, caregiver, audit trail, local cache, encrypted sync, consent, export, notification.

For iOS only? For Android only? unclear; parser should capture platform if visible.

- DS-042-2: App shall process heart rate data and preserve timestamp/timezone context.
- For iOS only? For Android only? unclear; parser should capture platform if visible.

ID	Requirement / Test / Step	Area	Risk	Acceptance / Expected Result	Status
DS-042-01	Show trend chart for 7/30/90 days	Backend API	Not ranked	TBD by clinical SME - intentionally incomplete.	Pending QA
DS_42.2	Display current BPM with timestamp	Backend API	Not ranked	System shall complete the action without data loss.	Pending QA
DS-042-03	Generate high/low heart-rate alert	Android/iOS	Medium	Result must be visible in mobile UI within expected time.	Pending QA
DS-042-04	Show trend chart for 7/30/90 days	Cloud Sync	High	User shall receive clear error message on failure.	Draft
DS_42.5	Display current BPM with timestamp	Backend API	Not ranked	Data shall be encrypted in transit and at rest.	In Review

DS Section 43 - Repeated requirement block

REVIEW-LATER

This page belongs to Design Specification for the Heart Rate Monitor mobile application. The application supports Android and iOS users, paired heart-rate sensors, manual BPM entry, trend review, alerts, data export, and controlled access. This block intentionally includes uneven numbering, repeated labels, partial owner fields, and mixed wording so an AI traceability/RAG engine can be tested against non-perfect documents. Page marker: DS-043. Owner: Clinical SME. Important terms: BPM, sensor pairing, low alert, high alert, caregiver, audit trail, local cache, encrypted sync, consent, export, notification.

38. Unnumbered note - check sensor disconnect behavior during active monitoring.

Unnumbered note - check sensor disconnect behavior during active monitoring.

ID	Requirement / Test / Step	Area	Risk	Acceptance / Expected Result	Status
DS-043-01	Sync readings to cloud API	iOS	Not ranked	System shall complete the action without data loss.	Rejected once
DS-043-02	Sync readings to cloud API	Android	High	TBD by clinical SME - intentionally incomplete.	Rejected once
DS-043-03	Export readings as PDF/CSV	Backend API	High	User shall receive clear error message on failure.	Pending QA
DS-043-04	Battery and Bluetooth warning	Android/iOS	Low	User shall receive clear error message on failure.	Draft
DS-043-05	Notification retry and suppression	Android/iOS	High	Audit/event record shall be generated where applicable.	Pending QA

DS Section 44 - Risk and control mapping

This page belongs to Design Specification for the Heart Rate Monitor mobile application. The application supports Android and iOS users, paired heart-rate sensors, manual BPM entry, trend review, alerts, data export, and controlled access. This block intentionally includes uneven numbering, repeated labels, partial owner fields, and mixed wording so an AI traceability/RAG engine can be tested against non-perfect documents. Page marker: DS-044. Owner: Product. Important terms: BPM, sensor pairing, low alert, high alert, caregiver, audit trail, local cache, encrypted sync, consent, export, notification.

DS-044-1: App shall process heart rate data and preserve timestamp/timezone context.

- Unnumbered note - check sensor disconnect behavior during active monitoring.
39. For iOS only? For Android only? unclear; parser should capture platform if visible.
- Old requirement copied from web portal project - review if still valid.

ID	Requirement / Test / Step	Area	Risk	Acceptance / Expected Result	Status
DS_44.1	Sync readings to cloud API	Local DB	Not ranked	TBD by clinical SME - intentionally incomplete.	Draft
DS-044-02	Store readings securely on device	Bluetooth Sensor	Not ranked	Audit/event record shall be generated where applicable.	Pending QA
DS-044-03	Manual heart rate entry when device unavailable	Local DB	Low	Data shall be encrypted in transit and at rest.	Draft
DS-044-04	Validate sensor data range	Bluetooth Sensor	Low	TBD by clinical SME - intentionally incomplete.	Approved?

DS Section 45 - Alert workflow

This page belongs to Design Specification for the Heart Rate Monitor mobile application. The application supports Android and iOS users, paired heart-rate sensors, manual BPM entry, trend review, alerts, data export, and controlled access. This block intentionally includes uneven numbering, repeated labels, partial owner fields, and mixed wording so an AI traceability/RAG engine can be tested against non-perfect documents. Page marker: DS-045. Owner: Clinical SME. Important terms: BPM, sensor pairing, low alert, high alert, caregiver, audit trail, local cache, encrypted sync, consent, export, notification.

40. DS-045-1: App shall process heart rate data and preserve timestamp/timezone context.

- Unnumbered note - check sensor disconnect behavior during active monitoring.

ID	Requirement / Test / Step	Area	Risk	Acceptance / Expected Result	Status
DS-045-01	Display current BPM with timestamp	Android/iOS	Low	Data shall be encrypted in transit and at rest.	Rejected once
DS-045-02	Support caregiver view	iOS	High	Audit/event record shall be generated where applicable.	Approved?
DS-045-03	Validate sensor data range	iOS	Low	Data shall be encrypted in transit and at rest.	Rejected once
DS-045-04	Capture heart rate from paired device	Bluetooth Sensor	TBD	User shall receive clear error message on failure.	Draft
DS-045-05	Handle offline mode	Android/iOS	Not ranked	System shall complete the action without data loss.	Approved?

MIXED TRACEABILITY LINE: URS-00X -> FRS?? -> DS component maybe -> IQ/OQ/PQ evidence TBD. This line is intentionally noisy.

DS Section 46 - Exception handling

This page belongs to Design Specification for the Heart Rate Monitor mobile application. The application supports Android and iOS users, paired heart-rate sensors, manual BPM entry, trend review, alerts, data export, and controlled access. This block intentionally includes uneven numbering, repeated labels, partial owner fields, and mixed wording so an AI traceability/RAG engine can be tested against non-perfect documents. Page marker: DS-046. Owner: Clinical SME. Important terms: BPM, sensor pairing, low alert, high alert, caregiver, audit trail, local cache, encrypted sync, consent, export, notification.

Unnumbered note - check sensor disconnect behavior during active monitoring.

41. Unnumbered note - check sensor disconnect behavior during active monitoring.

- DS-046-3: App shall process heart rate data and preserve timestamp/timezone context.

ID	Requirement / Test / Step	Area	Risk	Acceptance / Expected Result	Status
DS-046-01	Display current BPM with timestamp	Cloud Sync	High	TBD by clinical SME - intentionally incomplete.	Draft
DS_46.2	Support caregiver view	Cloud Sync	Low	User shall receive clear error message on failure.	Approved?
DS-046-03	Capture heart rate from paired device	Local DB	Medium	TBD by clinical SME - intentionally incomplete.	Pending QA
DS-046-04	Notification retry and suppression	Notification Service	TBD	Data shall be encrypted in transit and at rest.	In Review
DS-046-05	Generate high/low heart-rate alert	Android/iOS	TBD	Result must be visible in mobile UI within expected time.	Rejected once

DS Section 47 - UI behavior

This page belongs to Design Specification for the Heart Rate Monitor mobile application. The application supports Android and iOS users, paired heart-rate sensors, manual BPM entry, trend review, alerts, data export, and controlled access. This block intentionally includes uneven numbering, repeated labels, partial owner fields, and mixed wording so an AI traceability/RAG engine can be tested against non-perfect documents. Page marker: DS-047. Owner: Mobile Lead. Important terms: BPM, sensor pairing, low alert, high alert, caregiver, audit trail, local cache, encrypted sync, consent, export, notification.

42. Old requirement copied from web portal project - review if still valid.

Unnumbered note - check sensor disconnect behavior during active monitoring.

DS-047-3: App shall process heart rate data and preserve timestamp/timezone context.

For iOS only? For Android only? unclear; parser should capture platform if visible.

ID	Requirement / Test / Step	Area	Risk	Acceptance / Expected Result	Status
DS_47.1	Store readings securely on device	Bluetooth Sensor	Medium	Audit/event record shall be generated where applicable.	Rejected once
DS-047-02	Support caregiver view	Local DB	Medium	User shall receive clear error message on failure.	Rejected once
DS_47.3	Audit user actions	iOS	Low	TBD by clinical SME - intentionally incomplete.	Draft
DS-047-04	Validate sensor data range	Android/iOS	Low	User shall receive clear error message on failure.	In Review

DS Section 48 - Test condition table

This page belongs to Design Specification for the Heart Rate Monitor mobile application. The application supports Android and iOS users, paired heart-rate sensors, manual BPM entry, trend review, alerts, data export, and controlled access. This block intentionally includes uneven numbering, repeated labels, partial owner fields, and mixed wording so an AI traceability/RAG engine can be tested against non-perfect documents. Page marker: DS-048. Owner: Clinical SME. Important terms: BPM, sensor pairing, low alert, high alert, caregiver, audit trail, local cache, encrypted sync, consent, export, notification.

- Acceptance: expected result to be confirmed by QA; evidence screenshot may be attached later.
- DS-048-2: App shall process heart rate data and preserve timestamp/timezone context.

ID	Requirement / Test / Step	Area	Risk	Acceptance / Expected Result	Status
DS-048-01	Display current BPM with timestamp	Android	High	System shall complete the action without data loss.	Draft
DS-048-02	Support caregiver view	Android/iOS	Low	Audit/event record shall be generated where applicable.	Need SME
DS-048-03	Handle offline mode	Android	Low	Result must be visible in mobile UI within expected time.	Pending QA

DS Section 49 - API behavior

This page belongs to Design Specification for the Heart Rate Monitor mobile application. The application supports Android and iOS users, paired heart-rate sensors, manual BPM entry, trend review, alerts, data export, and controlled access. This block intentionally includes uneven numbering, repeated labels, partial owner fields, and mixed wording so an AI traceability/RAG engine can be tested against non-perfect documents. Page marker: DS-049. Owner: Clinical SME. Important terms: BPM, sensor pairing, low alert, high alert, caregiver, audit trail, local cache, encrypted sync, consent, export, notification.

- Old requirement copied from web portal project - review if still valid.

43. Old requirement copied from web portal project - review if still valid.

ID	Requirement / Test / Step	Area	Risk	Acceptance / Expected Result	Status
DS-049-01	Show trend chart for 7/30/90 days	Cloud Sync	High	System shall complete the action without data loss.	Rejected once
DS-049-02	Notification retry and suppression	Android	TBD	System shall complete the action without data loss.	In Review
DS-049-03	Notification retry and suppression	Android/iOS	Not ranked	Data shall be encrypted in transit and at rest.	Pending QA

DS Section 50 - Security and privacy

This page belongs to Design Specification for the Heart Rate Monitor mobile application. The application supports Android and iOS users, paired heart-rate sensors, manual BPM entry, trend review, alerts, data export, and controlled access. This block intentionally includes uneven numbering, repeated labels, partial owner fields, and mixed wording so an AI traceability/RAG engine can be tested against non-perfect documents. Page marker: DS-050. Owner: Mobile Lead. Important terms: BPM, sensor pairing, low alert, high alert, caregiver, audit trail, local cache, encrypted sync, consent, export, notification.

44. DS-050-1: App shall process heart rate data and preserve timestamp/timezone context.

- Acceptance: expected result to be confirmed by QA; evidence screenshot may be attached later.

45. DS-050-3: App shall process heart rate data and preserve timestamp/timezone context.

46. Unnumbered note - check sensor disconnect behavior during active monitoring.

ID	Requirement / Test / Step	Area	Risk	Acceptance / Expected Result	Status
DS_50.1	Audit user actions	Bluetooth Sensor	TBD	TBD by clinical SME - intentionally incomplete.	Rejected once
DS-050-02	Biometric login option	Notification Service	Medium	User shall receive clear error message on failure.	In Review
DS-050-03	Capture heart rate from paired device	iOS	Low	Data shall be encrypted in transit and at rest.	Draft
DS-050-04	Biometric login option	Bluetooth Sensor	High	Data shall be encrypted in transit and at rest.	Need SME
DS-050-05	Audit user actions	Bluetooth Sensor	Medium	User shall receive clear error message on failure.	Draft

DS Section 51 - Open questions

This page belongs to Design Specification for the Heart Rate Monitor mobile application. The application supports Android and iOS users, paired heart-rate sensors, manual BPM entry, trend review, alerts, data export, and controlled access. This block intentionally includes uneven numbering, repeated labels, partial owner fields, and mixed wording so an AI traceability/RAG engine can be tested against non-perfect documents. Page marker: DS-051. Owner: Clinical SME. Important terms: BPM, sensor pairing, low alert, high alert, caregiver, audit trail, local cache, encrypted sync, consent, export, notification.

- Old requirement copied from web portal project - review if still valid.

For iOS only? For Android only? unclear; parser should capture platform if visible.

ID	Requirement / Test / Step	Area	Risk	Acceptance / Expected Result	Status
DS-051-01	Generate high/low heart-rate alert	Android	Low	Result must be visible in mobile UI within expected time.	Rejected once
DS-051-02	Validate sensor data range	Local DB	Low	System shall complete the action without data loss.	Need SME
DS-051-03	Show trend chart for 7/30/90 days	iOS	Medium	User shall receive clear error message on failure.	Draft
DS-051-04	Display current BPM with timestamp	Notification Service	High	User shall receive clear error message on failure.	Draft

DS Section 52 - Data handling notes

This page belongs to Design Specification for the Heart Rate Monitor mobile application. The application supports Android and iOS users, paired heart-rate sensors, manual BPM entry, trend review, alerts, data export, and controlled access. This block intentionally includes uneven numbering, repeated labels, partial owner fields, and mixed wording so an AI traceability/RAG engine can be tested against non-perfect documents. Page marker: DS-052. Owner: TBD. Important terms: BPM, sensor pairing, low alert, high alert, caregiver, audit trail, local cache, encrypted sync, consent, export, notification.

- 47. Acceptance: expected result to be confirmed by QA; evidence screenshot may be attached later.
- 48. DS-052-2: App shall process heart rate data and preserve timestamp/timezone context.

- For iOS only? For Android only? unclear; parser should capture platform if visible.

Old requirement copied from web portal project - review if still valid.

ID	Requirement / Test / Step	Area	Risk	Acceptance / Expected Result	Status
DS-052-01	Export readings as PDF/CSV	Android/iOS	TBD	User shall receive clear error message on failure.	Need SME
DS_52.2	Display current BPM with timestamp	Bluetooth Sensor	High	TBD by clinical SME - intentionally incomplete.	Need SME
DS-052-03	Validate sensor data range	Bluetooth Sensor	Low	Result must be visible in mobile UI within expected time.	Draft
DS-052-04	Generate high/low heart-rate alert	Local DB	Low	Data shall be encrypted in transit and at rest.	In Review

Random inserted note: duplicate title appears again below.

DS Section 52 - Data handling notes

DS Section 53 - Open questions

missing owner

This page belongs to Design Specification for the Heart Rate Monitor mobile application. The application supports Android and iOS users, paired heart-rate sensors, manual BPM entry, trend review, alerts, data export, and controlled access. This block intentionally includes uneven numbering, repeated labels, partial owner fields, and mixed wording so an AI traceability/RAG engine can be tested against non-perfect documents. Page marker: DS-053. Owner: Validation. Important terms: BPM, sensor pairing, low alert, high alert, caregiver, audit trail, local cache, encrypted sync, consent, export, notification.

DS-053-1: App shall process heart rate data and preserve timestamp/timezone context.

49. Unnumbered note - check sensor disconnect behavior during active monitoring.

ID	Requirement / Test / Step	Area	Risk	Acceptance / Expected Result	Status
DS-053-01	Capture heart rate from paired device	Backend API	Not ranked	User shall receive clear error message on failure.	In Review
DS-053-02	Capture heart rate from paired device	Cloud Sync	High	Audit/event record shall be generated where applicable.	Pending QA
DS-053-03	Notification retry and suppression	Cloud Sync	High	User shall receive clear error message on failure.	Rejected once
DS-053-04	Notification retry and suppression	Backend API	Low	TBD by clinical SME - intentionally incomplete.	Approved?

DS Section 54 - Security and privacy

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This page belongs to Design Specification for the Heart Rate Monitor mobile application. The application supports Android and iOS users, paired heart-rate sensors, manual BPM entry, trend review, alerts, data export, and controlled access. This block intentionally includes uneven numbering, repeated labels, partial owner fields, and mixed wording so an AI traceability/RAG engine can be tested against non-perfect documents. Page marker: DS-054. Owner: Product. Important terms: BPM, sensor pairing, low alert, high alert, caregiver, audit trail, local cache, encrypted sync, consent, export, notification.

For iOS only? For Android only? unclear; parser should capture platform if visible.

- Unnumbered note - check sensor disconnect behavior during active monitoring.

ID	Requirement / Test / Step	Area	Risk	Acceptance / Expected Result	Status
DS-054-01	Handle offline mode	iOS	Not ranked	System shall complete the action without data loss.	Approved?
DS_54.2	Export readings as PDF/CSV	Android	TBD	Data shall be encrypted in transit and at rest.	Pending QA
DS-054-03	Export readings as PDF/CSV	Notification Service	Not ranked	Result must be visible in mobile UI within expected time.	Approved?
DS_54.4	Show trend chart for 7/30/90 days	Backend API	Low	Result must be visible in mobile UI within expected time.	Need SME
DS_54.5	Consent capture and privacy notice	Local DB	Medium	TBD by clinical SME - intentionally incomplete.	In Review

MIXED TRACEABILITY LINE: URS-00X -> FRS?? -> DS component maybe -> IQ/OQ/PQ evidence TBD. This line is intentionally noisy.

DS Section 55 - Alert workflow

This page belongs to Design Specification for the Heart Rate Monitor mobile application. The application supports Android and iOS users, paired heart-rate sensors, manual BPM entry, trend review, alerts, data export, and controlled access. This block intentionally includes uneven numbering, repeated labels, partial owner fields, and mixed wording so an AI traceability/RAG engine can be tested against non-perfect documents. Page marker: DS-055. Owner: Clinical SME. Important terms: BPM, sensor pairing, low alert, high alert, caregiver, audit trail, local cache, encrypted sync, consent, export, notification.

- For iOS only? For Android only? unclear; parser should capture platform if visible.
- Unnumbered note - check sensor disconnect behavior during active monitoring.

50. DS-055-3: App shall process heart rate data and preserve timestamp/timezone context.

Old requirement copied from web portal project - review if still valid.

ID	Requirement / Test / Step	Area	Risk	Acceptance / Expected Result	Status
DS-055-01	Export readings as PDF/CSV	Android/iOS	Not ranked	Result must be visible in mobile UI within expected time.	Pending QA
DS-055-02	Show trend chart for 7/30/90 days	Android/iOS	Low	Result must be visible in mobile UI within expected time.	Draft
DS_55.3	Validate sensor data range	iOS	Medium	Result must be visible in mobile UI within expected time.	Rejected once
DS-055-04	Support caregiver view	Local DB	Not ranked	Result must be visible in mobile UI within expected time.	Need SME

DS Section 56 - Data handling notes

This page belongs to Design Specification for the Heart Rate Monitor mobile application. The application supports Android and iOS users, paired heart-rate sensors, manual BPM entry, trend review, alerts, data export, and controlled access. This block intentionally includes uneven numbering, repeated labels, partial owner fields, and mixed wording so an AI traceability/RAG engine can be tested against non-perfect documents. Page marker: DS-056. Owner: Clinical SME. Important terms: BPM, sensor pairing, low alert, high alert, caregiver, audit trail, local cache, encrypted sync, consent, export, notification.

- Unnumbered note - check sensor disconnect behavior during active monitoring.
- For iOS only? For Android only? unclear; parser should capture platform if visible.

51. Unnumbered note - check sensor disconnect behavior during active monitoring.

- Old requirement copied from web portal project - review if still valid.

ID	Requirement / Test / Step	Area	Risk	Acceptance / Expected Result	Status
DS-056-01	Handle offline mode	Bluetooth Sensor	TBD	Audit/event record shall be generated where applicable.	Need SME
DS-056-02	Audit user actions	Android	High	System shall complete the action without data loss.	Rejected once
DS-056-03	Capture heart rate from paired device	Backend API	Low	TBD by clinical SME - intentionally incomplete.	Pending QA
DS-056-04	Export readings as PDF/CSV	Android	Low	Data shall be encrypted in transit and at rest.	Rejected once

DS Section 57 - Risk and control mapping

This page belongs to Design Specification for the Heart Rate Monitor mobile application. The application supports Android and iOS users, paired heart-rate sensors, manual BPM entry, trend review, alerts, data export, and controlled access. This block intentionally includes uneven numbering, repeated labels, partial owner fields, and mixed wording so an AI traceability/RAG engine can be tested against non-perfect documents. Page marker: DS-057. Owner: Mobile Lead. Important terms: BPM, sensor pairing, low alert, high alert, caregiver, audit trail, local cache, encrypted sync, consent, export, notification.

Acceptance: expected result to be confirmed by QA; evidence screenshot may be attached later.

52. Old requirement copied from web portal project - review if still valid.

- Unnumbered note - check sensor disconnect behavior during active monitoring.
- DS-057-4: App shall process heart rate data and preserve timestamp/timezone context.

ID	Requirement / Test / Step	Area	Risk	Acceptance / Expected Result	Status
DS-057-01	Show trend chart for 7/30/90 days	Bluetooth Sensor	Medium	Data shall be encrypted in transit and at rest.	Approved?
DS-057-02	Consent capture and privacy notice	Local DB	Low	TBD by clinical SME - intentionally incomplete.	Draft
DS-057-03	Biometric login option	Cloud Sync	High	Data shall be encrypted in transit and at rest.	Rejected once
DS-057-04	Manual heart rate entry when device unavailable	Local DB	Low	Result must be visible in mobile UI within expected time.	In Review
DS-057-05	Show trend chart for 7/30/90 days	Notification Service	TBD	Result must be visible in mobile UI within expected time.	Draft

DS Section 58 - Open questions

This page belongs to Design Specification for the Heart Rate Monitor mobile application. The application supports Android and iOS users, paired heart-rate sensors, manual BPM entry, trend review, alerts, data export, and controlled access. This block intentionally includes uneven numbering, repeated labels, partial owner fields, and mixed wording so an AI traceability/RAG engine can be tested against non-perfect documents. Page marker: DS-058. Owner: QA. Important terms: BPM, sensor pairing, low alert, high alert, caregiver, audit trail, local cache, encrypted sync, consent, export, notification.

DS-058-1: App shall process heart rate data and preserve timestamp/timezone context.

Unnumbered note - check sensor disconnect behavior during active monitoring.

ID	Requirement / Test / Step	Area	Risk	Acceptance / Expected Result	Status
DS-058-01	Capture heart rate from paired device	iOS	Medium	TBD by clinical SME - intentionally incomplete.	Rejected once
DS-058-02	Store readings securely on device	Notification Service	Medium	User shall receive clear error message on failure.	Draft
DS-058-03	Handle offline mode	Android	High	Result must be visible in mobile UI within expected time.	Approved?

DS Section 59 - Security and privacy

This page belongs to Design Specification for the Heart Rate Monitor mobile application. The application supports Android and iOS users, paired heart-rate sensors, manual BPM entry, trend review, alerts, data export, and controlled access. This block intentionally includes uneven numbering, repeated labels, partial owner fields, and mixed wording so an AI traceability/RAG engine can be tested against non-perfect documents. Page marker: DS-059. Owner: Clinical SME. Important terms: BPM, sensor pairing, low alert, high alert, caregiver, audit trail, local cache, encrypted sync, consent, export, notification.

53. DS-059-1: App shall process heart rate data and preserve timestamp/timezone context.

- DS-059-2: App shall process heart rate data and preserve timestamp/timezone context.

ID	Requirement / Test / Step	Area	Risk	Acceptance / Expected Result	Status
DS-059-01	Support caregiver view	iOS	Low	Result must be visible in mobile UI within expected time.	Pending QA
DS-059-02	Support caregiver view	iOS	Medium	Audit/event record shall be generated where applicable.	Need SME
DS-059-03	Manual heart rate entry when device unavailable	Notification Service	Medium	Result must be visible in mobile UI within expected time.	Approved?
DS-059-04	Capture heart rate from paired device	Android/iOS	Low	Data shall be encrypted in transit and at rest.	In Review
DS-059-05	Handle offline mode	Android/iOS	Low	System shall complete the action without data loss.	Draft

DS Section 60 - Test condition table

This page belongs to Design Specification for the Heart Rate Monitor mobile application. The application supports Android and iOS users, paired heart-rate sensors, manual BPM entry, trend review, alerts, data export, and controlled access. This block intentionally includes uneven numbering, repeated labels, partial owner fields, and mixed wording so an AI traceability/RAG engine can be tested against non-perfect documents. Page marker: DS-060. Owner: Mobile Lead. Important terms: BPM, sensor pairing, low alert, high alert, caregiver, audit trail, local cache, encrypted sync, consent, export, notification.

Acceptance: expected result to be confirmed by QA; evidence screenshot may be attached later.

54. Old requirement copied from web portal project - review if still valid.

- DS-060-3: App shall process heart rate data and preserve timestamp/timezone context.
- Acceptance: expected result to be confirmed by QA; evidence screenshot may be attached later.

ID	Requirement / Test / Step	Area	Risk	Acceptance / Expected Result	Status
DS_60.1	Display current BPM with timestamp	Android	High	Data shall be encrypted in transit and at rest.	Draft
DS_60.2	Export readings as PDF/CSV	Android	Not ranked	Data shall be encrypted in transit and at rest.	Rejected once
DS-060-03	Notification retry and suppression	iOS	Not ranked	User shall receive clear error message on failure.	Rejected once
DS_60.4	Audit user actions	Backend API	Not ranked	System shall complete the action without data loss.	In Review

DS Section 61 - Bluetooth edge cases

This page belongs to Design Specification for the Heart Rate Monitor mobile application. The application supports Android and iOS users, paired heart-rate sensors, manual BPM entry, trend review, alerts, data export, and controlled access. This block intentionally includes uneven numbering, repeated labels, partial owner fields, and mixed wording so an AI traceability/RAG engine can be tested against non-perfect documents. Page marker: DS-061. Owner: Mobile Lead. Important terms: BPM, sensor pairing, low alert, high alert, caregiver, audit trail, local cache, encrypted sync, consent, export, notification.

Unnumbered note - check sensor disconnect behavior during active monitoring.

- Unnumbered note - check sensor disconnect behavior during active monitoring.

For iOS only? For Android only? unclear; parser should capture platform if visible.

55. DS-061-4: App shall process heart rate data and preserve timestamp/timezone context.

ID	Requirement / Test / Step	Area	Risk	Acceptance / Expected Result	Status
DS-061-01	Manual heart rate entry when device unavailable	Notification Service	Low	Result must be visible in mobile UI within expected time.	Rejected once
DS-061-02	Export readings as PDF/CSV	Backend API	High	Result must be visible in mobile UI within expected time.	Rejected once
DS-061-03	Capture heart rate from paired device	Local DB	Medium	TBD by clinical SME - intentionally incomplete.	In Review
DS_61.4	Generate high/low heart-rate alert	Local DB	TBD	System shall complete the action without data loss.	Draft

DS Section 62 - Repeated requirement block

old text retained

This page belongs to Design Specification for the Heart Rate Monitor mobile application. The application supports Android and iOS users, paired heart-rate sensors, manual BPM entry, trend review, alerts, data export, and controlled access. This block intentionally includes uneven numbering, repeated labels, partial owner fields, and mixed wording so an AI traceability/RAG engine can be tested against non-perfect documents. Page marker: DS-062. Owner: Clinical SME. Important terms: BPM, sensor pairing, low alert, high alert, caregiver, audit trail, local cache, encrypted sync, consent, export, notification.

56. Old requirement copied from web portal project - review if still valid.

Acceptance: expected result to be confirmed by QA; evidence screenshot may be attached later.

Acceptance: expected result to be confirmed by QA; evidence screenshot may be attached later.

57. For iOS only? For Android only? unclear; parser should capture platform if visible.

ID	Requirement / Test / Step	Area	Risk	Acceptance / Expected Result	Status
DS-062-01	Generate high/low heart-rate alert	Backend API	TBD	System shall complete the action without data loss.	Approved?
DS-062-02	Audit user actions	Bluetooth Sensor	Not ranked	TBD by clinical SME - intentionally incomplete.	Approved?
DS-062-03	Support caregiver view	iOS	Medium	User shall receive clear error message on failure.	In Review
DS_62.4	Biometric login option	Android	Medium	TBD by clinical SME - intentionally incomplete.	Pending QA
DS_62.5	Consent capture and privacy notice	Local DB	TBD	Data shall be encrypted in transit and at rest.	Draft

DS Section 63 - Data handling notes

This page belongs to Design Specification for the Heart Rate Monitor mobile application. The application supports Android and iOS users, paired heart-rate sensors, manual BPM entry, trend review, alerts, data export, and controlled access. This block intentionally includes uneven numbering, repeated labels, partial owner fields, and mixed wording so an AI traceability/RAG engine can be tested against non-perfect documents. Page marker: DS-063. Owner: Mobile Lead. Important terms: BPM, sensor pairing, low alert, high alert, caregiver, audit trail, local cache, encrypted sync, consent, export, notification.

DS-063-1: App shall process heart rate data and preserve timestamp/timezone context.

Old requirement copied from web portal project - review if still valid.

Unnumbered note - check sensor disconnect behavior during active monitoring.

- Acceptance: expected result to be confirmed by QA; evidence screenshot may be attached later.

ID	Requirement / Test / Step	Area	Risk	Acceptance / Expected Result	Status
DS-063-01	Show trend chart for 7/30/90 days	Android/iOS	Low	Audit/event record shall be generated where applicable.	Need SME
DS-063-02	Support caregiver view	Cloud Sync	TBD	System shall complete the action without data loss.	Approved?
DS-063-03	Display current BPM with timestamp	Local DB	TBD	User shall receive clear error message on failure.	Rejected once

MIXED TRACEABILITY LINE: URS-00X -> FRS?? -> DS component maybe -> IQ/OQ/PQ evidence TBD. This line is intentionally noisy.

DS Section 64 - Exception handling

This page belongs to Design Specification for the Heart Rate Monitor mobile application. The application supports Android and iOS users, paired heart-rate sensors, manual BPM entry, trend review, alerts, data export, and controlled access. This block intentionally includes uneven numbering, repeated labels, partial owner fields, and mixed wording so an AI traceability/RAG engine can be tested against non-perfect documents. Page marker: DS-064. Owner: Validation. Important terms: BPM, sensor pairing, low alert, high alert, caregiver, audit trail, local cache, encrypted sync, consent, export, notification.

- DS-064-1: App shall process heart rate data and preserve timestamp/timezone context.
- Unnumbered note - check sensor disconnect behavior during active monitoring.
- For iOS only? For Android only? unclear; parser should capture platform if visible.

ID	Requirement / Test / Step	Area	Risk	Acceptance / Expected Result	Status
DS-064-01	Sync readings to cloud API	Local DB	TBD	System shall complete the action without data loss.	Draft
DS-064-02	Display current BPM with timestamp	Backend API	Not ranked	Data shall be encrypted in transit and at rest.	Pending QA
DS-064-03	Manual heart rate entry when device unavailable	Cloud Sync	Medium	Audit/event record shall be generated where applicable.	Draft

DS Section 65 - Data handling notes

This page belongs to Design Specification for the Heart Rate Monitor mobile application. The application supports Android and iOS users, paired heart-rate sensors, manual BPM entry, trend review, alerts, data export, and controlled access. This block intentionally includes uneven numbering, repeated labels, partial owner fields, and mixed wording so an AI traceability/RAG engine can be tested against non-perfect documents. Page marker: DS-065. Owner: Clinical SME. Important terms: BPM, sensor pairing, low alert, high alert, caregiver, audit trail, local cache, encrypted sync, consent, export, notification.

Acceptance: expected result to be confirmed by QA; evidence screenshot may be attached later.

58. Unnumbered note - check sensor disconnect behavior during active monitoring.

- For iOS only? For Android only? unclear; parser should capture platform if visible.

ID	Requirement / Test / Step	Area	Risk	Acceptance / Expected Result	Status
DS_65.1	Display current BPM with timestamp	iOS	High	Result must be visible in mobile UI within expected time.	Draft
DS-065-02	Handle offline mode	Local DB	Not ranked	System shall complete the action without data loss.	Approved?
DS-065-03	Notification retry and suppression	Android	Not ranked	Audit/event record shall be generated where applicable.	Draft

Random inserted note: duplicate title appears again below.

DS Section 65 - Data handling notes

DS Section 66 - Requirement details

This page belongs to Design Specification for the Heart Rate Monitor mobile application. The application supports Android and iOS users, paired heart-rate sensors, manual BPM entry, trend review, alerts, data export, and controlled access. This block intentionally includes uneven numbering, repeated labels, partial owner fields, and mixed wording so an AI traceability/RAG engine can be tested against non-perfect documents. Page marker: DS-066. Owner: Product. Important terms: BPM, sensor pairing, low alert, high alert, caregiver, audit trail, local cache, encrypted sync, consent, export, notification.

DS-066-1: App shall process heart rate data and preserve timestamp/timezone context.

59. For iOS only? For Android only? unclear; parser should capture platform if visible.

ID	Requirement / Test / Step	Area	Risk	Acceptance / Expected Result	Status
DS-066-01	Store readings securely on device	Notification Service	Low	Result must be visible in mobile UI within expected time.	In Review
DS-066-02	Audit user actions	Bluetooth Sensor	Medium	System shall complete the action without data loss.	Approved?
DS-066-03	Notification retry and suppression	Android/iOS	Not ranked	Result must be visible in mobile UI within expected time.	Pending QA
DS-066-04	Store readings securely on device	Cloud Sync	High	TBD by clinical SME - intentionally incomplete.	Need SME
DS-066-05	Notification retry and suppression	Bluetooth Sensor	Not ranked	Data shall be encrypted in transit and at rest.	In Review

DS Section 67 - Exception handling

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This page belongs to Design Specification for the Heart Rate Monitor mobile application. The application supports Android and iOS users, paired heart-rate sensors, manual BPM entry, trend review, alerts, data export, and controlled access. This block intentionally includes uneven numbering, repeated labels, partial owner fields, and mixed wording so an AI traceability/RAG engine can be tested against non-perfect documents. Page marker: DS-067. Owner: Validation. Important terms: BPM, sensor pairing, low alert, high alert, caregiver, audit trail, local cache, encrypted sync, consent, export, notification.

- DS-067-1: App shall process heart rate data and preserve timestamp/timezone context.

Acceptance: expected result to be confirmed by QA; evidence screenshot may be attached later.

ID	Requirement / Test / Step	Area	Risk	Acceptance / Expected Result	Status
DS_67.1	Manual heart rate entry when device unavailable	Local DB	High	Data shall be encrypted in transit and at rest.	Pending QA
DS_67.2	Battery and Bluetooth warning	Cloud Sync	Not ranked	TBD by clinical SME - intentionally incomplete.	Rejected once
DS_67.3	Store readings securely on device	Android	TBD	TBD by clinical SME - intentionally incomplete.	In Review
DS_67.4	Show trend chart for 7/30/90 days	Backend API	Low	Data shall be encrypted in transit and at rest.	Rejected once

DS Section 68 - Test condition table

This page belongs to Design Specification for the Heart Rate Monitor mobile application. The application supports Android and iOS users, paired heart-rate sensors, manual BPM entry, trend review, alerts, data export, and controlled access. This block intentionally includes uneven numbering, repeated labels, partial owner fields, and mixed wording so an AI traceability/RAG engine can be tested against non-perfect documents. Page marker: DS-068. Owner: QA. Important terms: BPM, sensor pairing, low alert, high alert, caregiver, audit trail, local cache, encrypted sync, consent, export, notification.

DS-068-1: App shall process heart rate data and preserve timestamp/timezone context.

- 60. Old requirement copied from web portal project - review if still valid.
- 61. Old requirement copied from web portal project - review if still valid.

Old requirement copied from web portal project - review if still valid.

ID	Requirement / Test / Step	Area	Risk	Acceptance / Expected Result	Status
DS-068-01	Support caregiver view	iOS	High	User shall receive clear error message on failure.	Approved?
DS-068-02	Handle offline mode	Notification Service	Not ranked	Audit/event record shall be generated where applicable.	Approved?
DS-068-03	Export readings as PDF/CSV	Cloud Sync	Low	Data shall be encrypted in transit and at rest.	Approved?

DS Section 69 - Traceability notes

RAG TEST CHUNK

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For iOS only? For Android only? unclear; parser should capture platform if visible.

- Acceptance: expected result to be confirmed by QA; evidence screenshot may be attached later.

ID	Requirement / Test / Step	Area	Risk	Acceptance / Expected Result	Status
DS-069-01	Audit user actions	Notification Service	Medium	Data shall be encrypted in transit and at rest.	Approved?
DS-069-02	Support caregiver view	Cloud Sync	Medium	User shall receive clear error message on failure.	Need SME
DS-069-03	Biometric login option	Notification Service	Low	User shall receive clear error message on failure.	Need SME
DS-069-04	Consent capture and privacy notice	iOS	High	User shall receive clear error message on failure.	Need SME

DS Section 70 - Risk and control mapping

This page belongs to Design Specification for the Heart Rate Monitor mobile application. The application supports Android and iOS users, paired heart-rate sensors, manual BPM entry, trend review, alerts, data export, and controlled access. This block intentionally includes uneven numbering, repeated labels, partial owner fields, and mixed wording so an AI traceability/RAG engine can be tested against non-perfect documents. Page marker: DS-070. Owner: Clinical SME. Important terms: BPM, sensor pairing, low alert, high alert, caregiver, audit trail, local cache, encrypted sync, consent, export, notification.

- DS-070-1: App shall process heart rate data and preserve timestamp/timezone context.

62. Unnumbered note - check sensor disconnect behavior during active monitoring.

DS-070-3: App shall process heart rate data and preserve timestamp/timezone context.

ID	Requirement / Test / Step	Area	Risk	Acceptance / Expected Result	Status
DS-070-01	Store readings securely on device	iOS	Low	Data shall be encrypted in transit and at rest.	In Review
DS-070-02	Consent capture and privacy notice	Android	Not ranked	Audit/event record shall be generated where applicable.	Approved?
DS-070-03	Export readings as PDF/CSV	Android/iOS	TBD	TBD by clinical SME - intentionally incomplete.	Need SME

DS Section 71 - Alert workflow

v0.1/v1.0 mixed

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DS-071-1: App shall process heart rate data and preserve timestamp/timezone context.

63. Old requirement copied from web portal project - review if still valid.

64. Acceptance: expected result to be confirmed by QA; evidence screenshot may be attached later.

ID	Requirement / Test / Step	Area	Risk	Acceptance / Expected Result	Status
DS-071-01	Biometric login option	Android	Not ranked	TBD by clinical SME - intentionally incomplete.	In Review
DS-071-02	Export readings as PDF/CSV	Android/iOS	Low	TBD by clinical SME - intentionally incomplete.	Rejected once
DS-071-03	Sync readings to cloud API	Bluetooth Sensor	Not ranked	Result must be visible in mobile UI within expected time.	Need SME
DS-071-04	Generate high/low heart-rate alert	Android	Low	Audit/event record shall be generated where applicable.	In Review
DS-071-05	Generate high/low heart-rate alert	Android	Not ranked	System shall complete the action without data loss.	Draft

DS Section 72 - Alert workflow

This page belongs to Design Specification for the Heart Rate Monitor mobile application. The application supports Android and iOS users, paired heart-rate sensors, manual BPM entry, trend review, alerts, data export, and controlled access. This block intentionally includes uneven numbering, repeated labels, partial owner fields, and mixed wording so an AI traceability/RAG engine can be tested against non-perfect documents. Page marker: DS-072. Owner: QA. Important terms: BPM, sensor pairing, low alert, high alert, caregiver, audit trail, local cache, encrypted sync, consent, export, notification.

DS-072-1: App shall process heart rate data and preserve timestamp/timezone context.

65. Old requirement copied from web portal project - review if still valid.

- DS-072-3: App shall process heart rate data and preserve timestamp/timezone context.

66. Old requirement copied from web portal project - review if still valid.

ID	Requirement / Test / Step	Area	Risk	Acceptance / Expected Result	Status
DS-072-01	Display current BPM with timestamp	iOS	TBD	System shall complete the action without data loss.	Draft
DS-072-02	Store readings securely on device	Notification Service	High	Audit/event record shall be generated where applicable.	Approved?
DS-072-03	Generate high/low heart-rate alert	Backend API	Not ranked	Result must be visible in mobile UI within expected time.	In Review
DS-072-04	Show trend chart for 7/30/90 days	Bluetooth Sensor	Not ranked	Audit/event record shall be generated where applicable.	Rejected once
DS_72.5	Display current BPM with timestamp	Local DB	Medium	Result must be visible in mobile UI within expected time.	Pending QA

MIXED TRACEABILITY LINE: URS-00X -> FRS?? -> DS component maybe -> IQ/OQ/PQ evidence TBD. This line is intentionally noisy.

DS Section 73 - Traceability notes

manual note

This page belongs to Design Specification for the Heart Rate Monitor mobile application. The application supports Android and iOS users, paired heart-rate sensors, manual BPM entry, trend review, alerts, data export, and controlled access. This block intentionally includes uneven numbering, repeated labels, partial owner fields, and mixed wording so an AI traceability/RAG engine can be tested against non-perfect documents. Page marker: DS-073. Owner: QA. Important terms: BPM, sensor pairing, low alert, high alert, caregiver, audit trail, local cache, encrypted sync, consent, export, notification.

67. Acceptance: expected result to be confirmed by QA; evidence screenshot may be attached later.

For iOS only? For Android only? unclear; parser should capture platform if visible.

- Acceptance: expected result to be confirmed by QA; evidence screenshot may be attached later.

68. Old requirement copied from web portal project - review if still valid.

ID	Requirement / Test / Step	Area	Risk	Acceptance / Expected Result	Status
DS_73.1	Capture heart rate from paired device	Android/iOS	Medium	TBD by clinical SME - intentionally incomplete.	In Review
DS_73.2	Biometric login option	Cloud Sync	Low	System shall complete the action without data loss.	Draft
DS-073-03	Support caregiver view	Backend API	Low	Result must be visible in mobile UI within expected time.	Pending QA

DS Section 74 - UI behavior

This page belongs to Design Specification for the Heart Rate Monitor mobile application. The application supports Android and iOS users, paired heart-rate sensors, manual BPM entry, trend review, alerts, data export, and controlled access. This block intentionally includes uneven numbering, repeated labels, partial owner fields, and mixed wording so an AI traceability/RAG engine can be tested against non-perfect documents. Page marker: DS-074. Owner: Clinical SME. Important terms: BPM, sensor pairing, low alert, high alert, caregiver, audit trail, local cache, encrypted sync, consent, export, notification.

Old requirement copied from web portal project - review if still valid.

- Old requirement copied from web portal project - review if still valid.

Old requirement copied from web portal project - review if still valid.

69. Old requirement copied from web portal project - review if still valid.

ID	Requirement / Test / Step	Area	Risk	Acceptance / Expected Result	Status
DS_74.1	Notification retry and suppression	Backend API	Low	Audit/event record shall be generated where applicable.	Rejected once
DS_74.2	Audit user actions	Android/iOS	Medium	Data shall be encrypted in transit and at rest.	Rejected once
DS-074-03	Generate high/low heart-rate alert	Bluetooth Sensor	High	Result must be visible in mobile UI within expected time.	Approved?

DS Section 75 - Mobile platform variation

This page belongs to Design Specification for the Heart Rate Monitor mobile application. The application supports Android and iOS users, paired heart-rate sensors, manual BPM entry, trend review, alerts, data export, and controlled access. This block intentionally includes uneven numbering, repeated labels, partial owner fields, and mixed wording so an AI traceability/RAG engine can be tested against non-perfect documents. Page marker: DS-075. Owner: Mobile Lead. Important terms: BPM, sensor pairing, low alert, high alert, caregiver, audit trail, local cache, encrypted sync, consent, export, notification.

70. Acceptance: expected result to be confirmed by QA; evidence screenshot may be attached later.

71. Acceptance: expected result to be confirmed by QA; evidence screenshot may be attached later.

ID	Requirement / Test / Step	Area	Risk	Acceptance / Expected Result	Status
DS-075-01	Sync readings to cloud API	iOS	Low	Result must be visible in mobile UI within expected time.	Draft
DS-075-02	Notification retry and suppression	Android	Medium	Audit/event record shall be generated where applicable.	Draft
DS_75.3	Store readings securely on device	Android	TBD	TBD by clinical SME - intentionally incomplete.	Need SME
DS_75.4	Notification retry and suppression	Local DB	Low	Data shall be encrypted in transit and at rest.	In Review

DS Section 76 - Mobile platform variation

TBD

This page belongs to Design Specification for the Heart Rate Monitor mobile application. The application supports Android and iOS users, paired heart-rate sensors, manual BPM entry, trend review, alerts, data export, and controlled access. This block intentionally includes uneven numbering, repeated labels, partial owner fields, and mixed wording so an AI traceability/RAG engine can be tested against non-perfect documents. Page marker: DS-076. Owner: Clinical SME. Important terms: BPM, sensor pairing, low alert, high alert, caregiver, audit trail, local cache, encrypted sync, consent, export, notification.

72. DS-076-1: App shall process heart rate data and preserve timestamp/timezone context.

Old requirement copied from web portal project - review if still valid.

ID	Requirement / Test / Step	Area	Risk	Acceptance / Expected Result	Status
DS-076-01	Store readings securely on device	Notification Service	Medium	User shall receive clear error message on failure.	In Review
DS-076-02	Battery and Bluetooth warning	Android/iOS	Not ranked	Audit/event record shall be generated where applicable.	Approved?
DS-076-03	Manual heart rate entry when device unavailable	Cloud Sync	High	TBD by clinical SME - intentionally incomplete.	Draft

DS Section 77 - Repeated requirement block

This page belongs to Design Specification for the Heart Rate Monitor mobile application. The application supports Android and iOS users, paired heart-rate sensors, manual BPM entry, trend review, alerts, data export, and controlled access. This block intentionally includes uneven numbering, repeated labels, partial owner fields, and mixed wording so an AI traceability/RAG engine can be tested against non-perfect documents. Page marker: DS-077. Owner: QA. Important terms: BPM, sensor pairing, low alert, high alert, caregiver, audit trail, local cache, encrypted sync, consent, export, notification.

Acceptance: expected result to be confirmed by QA; evidence screenshot may be attached later.

Unnumbered note - check sensor disconnect behavior during active monitoring.

73. Unnumbered note - check sensor disconnect behavior during active monitoring.

DS-077-4: App shall process heart rate data and preserve timestamp/timezone context.

ID	Requirement / Test / Step	Area	Risk	Acceptance / Expected Result	Status
DS-077-01	Notification retry and suppression	Notification Service	TBD	Data shall be encrypted in transit and at rest.	Pending QA
DS-077-02	Generate high/low heart-rate alert	Backend API	Low	System shall complete the action without data loss.	Approved?
DS-077-03	Sync readings to cloud API	Android/iOS	High	Result must be visible in mobile UI within expected time.	In Review
DS-077-04	Battery and Bluetooth warning	Cloud Sync	High	User shall receive clear error message on failure.	Rejected once
DS-077-05	Consent capture and privacy notice	Notification Service	Not ranked	System shall complete the action without data loss.	Rejected once

DS Section 78 - Exception handling

This page belongs to Design Specification for the Heart Rate Monitor mobile application. The application supports Android and iOS users, paired heart-rate sensors, manual BPM entry, trend review, alerts, data export, and controlled access. This block intentionally includes uneven numbering, repeated labels, partial owner fields, and mixed wording so an AI traceability/RAG engine can be tested against non-perfect documents. Page marker: DS-078. Owner: TBD. Important terms: BPM, sensor pairing, low alert, high alert, caregiver, audit trail, local cache, encrypted sync, consent, export, notification.

74. Old requirement copied from web portal project - review if still valid.

- Old requirement copied from web portal project - review if still valid.
- For iOS only? For Android only? unclear; parser should capture platform if visible.

ID	Requirement / Test / Step	Area	Risk	Acceptance / Expected Result	Status
DS-078-01	Consent capture and privacy notice	Android/iOS	TBD	User shall receive clear error message on failure.	Pending QA
DS-078-02	Consent capture and privacy notice	Backend API	Medium	TBD by clinical SME - intentionally incomplete.	Pending QA
DS-078-03	Consent capture and privacy notice	Cloud Sync	Medium	TBD by clinical SME - intentionally incomplete.	Draft
DS-078-04	Sync readings to cloud API	Notification Service	TBD	Audit/event record shall be generated where applicable.	Draft

Random inserted note: duplicate title appears again below.

DS Section 78 - Exception handling

DS Section 79 - Audit trail mapping

This page belongs to Design Specification for the Heart Rate Monitor mobile application. The application supports Android and iOS users, paired heart-rate sensors, manual BPM entry, trend review, alerts, data export, and controlled access. This block intentionally includes uneven numbering, repeated labels, partial owner fields, and mixed wording so an AI traceability/RAG engine can be tested against non-perfect documents. Page marker: DS-079. Owner: Validation. Important terms: BPM, sensor pairing, low alert, high alert, caregiver, audit trail, local cache, encrypted sync, consent, export, notification.

75. Unnumbered note - check sensor disconnect behavior during active monitoring.

- For iOS only? For Android only? unclear; parser should capture platform if visible.

ID	Requirement / Test / Step	Area	Risk	Acceptance / Expected Result	Status
DS_79.1	Display current BPM with timestamp	Android/iOS	High	System shall complete the action without data loss.	Rejected once
DS-079-02	Handle offline mode	iOS	High	TBD by clinical SME - intentionally incomplete.	Need SME
DS-079-03	Show trend chart for 7/30/90 days	Bluetooth Sensor	Not ranked	Data shall be encrypted in transit and at rest.	Need SME
DS-079-04	Manual heart rate entry when device unavailable	Notification Service	Medium	TBD by clinical SME - intentionally incomplete.	In Review

DS Section 80 - Traceability notes

This page belongs to Design Specification for the Heart Rate Monitor mobile application. The application supports Android and iOS users, paired heart-rate sensors, manual BPM entry, trend review, alerts, data export, and controlled access. This block intentionally includes uneven numbering, repeated labels, partial owner fields, and mixed wording so an AI traceability/RAG engine can be tested against non-perfect documents. Page marker: DS-080. Owner: Clinical SME. Important terms: BPM, sensor pairing, low alert, high alert, caregiver, audit trail, local cache, encrypted sync, consent, export, notification.

DS-080-1: App shall process heart rate data and preserve timestamp/timezone context.

- Acceptance: expected result to be confirmed by QA; evidence screenshot may be attached later.

76. Old requirement copied from web portal project - review if still valid.

Old requirement copied from web portal project - review if still valid.

ID	Requirement / Test / Step	Area	Risk	Acceptance / Expected Result	Status
DS-080-01	Generate high/low heart-rate alert	Android/iOS	Medium	Audit/event record shall be generated where applicable.	Need SME
DS_80.2	Generate high/low heart-rate alert	iOS	TBD	System shall complete the action without data loss.	Need SME
DS-080-03	Consent capture and privacy notice	Android/iOS	Not ranked	TBD by clinical SME - intentionally incomplete.	Draft

DS Section 81 - Open questions

This page belongs to Design Specification for the Heart Rate Monitor mobile application. The application supports Android and iOS users, paired heart-rate sensors, manual BPM entry, trend review, alerts, data export, and controlled access. This block intentionally includes uneven numbering, repeated labels, partial owner fields, and mixed wording so an AI traceability/RAG engine can be tested against non-perfect documents. Page marker: DS-081. Owner: Validation. Important terms: BPM, sensor pairing, low alert, high alert, caregiver, audit trail, local cache, encrypted sync, consent, export, notification.

Old requirement copied from web portal project - review if still valid.

Acceptance: expected result to be confirmed by QA; evidence screenshot may be attached later.

- DS-081-3: App shall process heart rate data and preserve timestamp/timezone context.

ID	Requirement / Test / Step	Area	Risk	Acceptance / Expected Result	Status
DS-081-01	Support caregiver view	Cloud Sync	Not ranked	User shall receive clear error message on failure.	Pending QA
DS_81.2	Generate high/low heart-rate alert	Bluetooth Sensor	Not ranked	Audit/event record shall be generated where applicable.	Pending QA
DS-081-03	Validate sensor data range	Local DB	Low	Result must be visible in mobile UI within expected time.	Approved?
DS-081-04	Show trend chart for 7/30/90 days	iOS	Medium	Result must be visible in mobile UI within expected time.	Approved?

MIXED TRACEABILITY LINE: URS-00X -> FRS?? -> DS component maybe -> IQ/OQ/PQ evidence TBD. This line is intentionally noisy.

DS Section 82 - Risk and control mapping

duplicate section

This page belongs to Design Specification for the Heart Rate Monitor mobile application. The application supports Android and iOS users, paired heart-rate sensors, manual BPM entry, trend review, alerts, data export, and controlled access. This block intentionally includes uneven numbering, repeated labels, partial owner fields, and mixed wording so an AI traceability/RAG engine can be tested against non-perfect documents. Page marker: DS-082. Owner: Product. Important terms: BPM, sensor pairing, low alert, high alert, caregiver, audit trail, local cache, encrypted sync, consent, export, notification.

77. Acceptance: expected result to be confirmed by QA; evidence screenshot may be attached later.

Acceptance: expected result to be confirmed by QA; evidence screenshot may be attached later.

ID	Requirement / Test / Step	Area	Risk	Acceptance / Expected Result	Status
DS-082-01	Biometric login option	Android	TBD	Data shall be encrypted in transit and at rest.	Draft
DS-082-02	Consent capture and privacy notice	iOS	Low	Data shall be encrypted in transit and at rest.	Rejected once
DS-082-03	Validate sensor data range	Notification Service	High	TBD by clinical SME - intentionally incomplete.	In Review
DS-082-04	Store readings securely on device	iOS	TBD	Result must be visible in mobile UI within expected time.	Approved?

DS Section 83 - Repeated requirement block

This page belongs to Design Specification for the Heart Rate Monitor mobile application. The application supports Android and iOS users, paired heart-rate sensors, manual BPM entry, trend review, alerts, data export, and controlled access. This block intentionally includes uneven numbering, repeated labels, partial owner fields, and mixed wording so an AI traceability/RAG engine can be tested against non-perfect documents. Page marker: DS-083. Owner: TBD. Important terms: BPM, sensor pairing, low alert, high alert, caregiver, audit trail, local cache, encrypted sync, consent, export, notification.

Acceptance: expected result to be confirmed by QA; evidence screenshot may be attached later.

78. Old requirement copied from web portal project - review if still valid.

ID	Requirement / Test / Step	Area	Risk	Acceptance / Expected Result	Status
DS-083-01	Manual heart rate entry when device unavailable	Bluetooth Sensor	Medium	Audit/event record shall be generated where applicable.	Rejected once
DS-083-02	Generate high/low heart-rate alert	Backend API	High	Result must be visible in mobile UI within expected time.	In Review
DS-083-03	Audit user actions	Android/iOS	TBD	User shall receive clear error message on failure.	Approved?

DS Section 84 - Traceability notes

This page belongs to Design Specification for the Heart Rate Monitor mobile application. The application supports Android and iOS users, paired heart-rate sensors, manual BPM entry, trend review, alerts, data export, and controlled access. This block intentionally includes uneven numbering, repeated labels, partial owner fields, and mixed wording so an AI traceability/RAG engine can be tested against non-perfect documents. Page marker: DS-084. Owner: Mobile Lead. Important terms: BPM, sensor pairing, low alert, high alert, caregiver, audit trail, local cache, encrypted sync, consent, export, notification.

- Unnumbered note - check sensor disconnect behavior during active monitoring.
- Acceptance: expected result to be confirmed by QA; evidence screenshot may be attached later.

ID	Requirement / Test / Step	Area	Risk	Acceptance / Expected Result	Status
DS_84.1	Display current BPM with timestamp	Android	Medium	TBD by clinical SME - intentionally incomplete.	In Review
DS-084-02	Show trend chart for 7/30/90 days	Android	Low	Result must be visible in mobile UI within expected time.	Approved?
DS-084-03	Sync readings to cloud API	iOS	Not ranked	TBD by clinical SME - intentionally incomplete.	Pending QA
DS-084-04	Validate sensor data range	iOS	TBD	System shall complete the action without data loss.	Approved?

DS Section 85 - Repeated requirement block

This page belongs to Design Specification for the Heart Rate Monitor mobile application. The application supports Android and iOS users, paired heart-rate sensors, manual BPM entry, trend review, alerts, data export, and controlled access. This block intentionally includes uneven numbering, repeated labels, partial owner fields, and mixed wording so an AI traceability/RAG engine can be tested against non-perfect documents. Page marker: DS-085. Owner: Mobile Lead. Important terms: BPM, sensor pairing, low alert, high alert, caregiver, audit trail, local cache, encrypted sync, consent, export, notification.

Old requirement copied from web portal project - review if still valid.

- For iOS only? For Android only? unclear; parser should capture platform if visible.

ID	Requirement / Test / Step	Area	Risk	Acceptance / Expected Result	Status
DS-085-01	Display current BPM with timestamp	Cloud Sync	Low	Result must be visible in mobile UI within expected time.	Need SME
DS-085-02	Export readings as PDF/CSV	Local DB	High	TBD by clinical SME - intentionally incomplete.	Rejected once
DS-085-03	Capture heart rate from paired device	Backend API	High	TBD by clinical SME - intentionally incomplete.	Pending QA

DS Section 86 - Evidence placeholder

This page belongs to Design Specification for the Heart Rate Monitor mobile application. The application supports Android and iOS users, paired heart-rate sensors, manual BPM entry, trend review, alerts, data export, and controlled access. This block intentionally includes uneven numbering, repeated labels, partial owner fields, and mixed wording so an AI traceability/RAG engine can be tested against non-perfect documents. Page marker: DS-086. Owner: Mobile Lead. Important terms: BPM, sensor pairing, low alert, high alert, caregiver, audit trail, local cache, encrypted sync, consent, export, notification.

79. Unnumbered note - check sensor disconnect behavior during active monitoring.

- Old requirement copied from web portal project - review if still valid.

Old requirement copied from web portal project - review if still valid.

80. Acceptance: expected result to be confirmed by QA; evidence screenshot may be attached later.

ID	Requirement / Test / Step	Area	Risk	Acceptance / Expected Result	Status
DS-086-01	Handle offline mode	Android	Low	Result must be visible in mobile UI within expected time.	Approved?
DS-086-02	Generate high/low heart-rate alert	Bluetooth Sensor	TBD	Result must be visible in mobile UI within expected time.	Approved?
DS-086-03	Support caregiver view	Cloud Sync	Not ranked	User shall receive clear error message on failure.	In Review
DS-086-04	Generate high/low heart-rate alert	iOS	Not ranked	System shall complete the action without data loss.	Draft
DS-086-05	Handle offline mode	Cloud Sync	Low	System shall complete the action without data loss.	Rejected once

DS Section 87 - Risk and control mapping

manual note

This page belongs to Design Specification for the Heart Rate Monitor mobile application. The application supports Android and iOS users, paired heart-rate sensors, manual BPM entry, trend review, alerts, data export, and controlled access. This block intentionally includes uneven numbering, repeated labels, partial owner fields, and mixed wording so an AI traceability/RAG engine can be tested against non-perfect documents. Page marker: DS-087. Owner: Clinical SME. Important terms: BPM, sensor pairing, low alert, high alert, caregiver, audit trail, local cache, encrypted sync, consent, export, notification.

DS-087-1: App shall process heart rate data and preserve timestamp/timezone context.

81. Old requirement copied from web portal project - review if still valid.

ID	Requirement / Test / Step	Area	Risk	Acceptance / Expected Result	Status
DS-087-01	Store readings securely on device	Bluetooth Sensor	Not ranked	Result must be visible in mobile UI within expected time.	Approved?
DS-087-02	Display current BPM with timestamp	Cloud Sync	High	Data shall be encrypted in transit and at rest.	Rejected once
DS-087-03	Manual heart rate entry when device unavailable	Android/iOS	Low	System shall complete the action without data loss.	Approved?

DS Section 88 - Security and privacy

OCR noisy line

This page belongs to Design Specification for the Heart Rate Monitor mobile application. The application supports Android and iOS users, paired heart-rate sensors, manual BPM entry, trend review, alerts, data export, and controlled access. This block intentionally includes uneven numbering, repeated labels, partial owner fields, and mixed wording so an AI traceability/RAG engine can be tested against non-perfect documents. Page marker: DS-088. Owner: Mobile Lead. Important terms: BPM, sensor pairing, low alert, high alert, caregiver, audit trail, local cache, encrypted sync, consent, export, notification.

- Old requirement copied from web portal project - review if still valid.
- For iOS only? For Android only? unclear; parser should capture platform if visible.
- For iOS only? For Android only? unclear; parser should capture platform if visible.

82. For iOS only? For Android only? unclear; parser should capture platform if visible.

ID	Requirement / Test / Step	Area	Risk	Acceptance / Expected Result	Status
DS-088-01	Generate high/low heart-rate alert	Local DB	Not ranked	Result must be visible in mobile UI within expected time.	Draft
DS-088-02	Biometric login option	Bluetooth Sensor	Low	Result must be visible in mobile UI within expected time.	Pending QA
DS-088-03	Audit user actions	iOS	Not ranked	System shall complete the action without data loss.	Pending QA
DS-088-04	Battery and Bluetooth warning	iOS	High	System shall complete the action without data loss.	Approved?

DS Section 89 - Requirement details

This page belongs to Design Specification for the Heart Rate Monitor mobile application. The application supports Android and iOS users, paired heart-rate sensors, manual BPM entry, trend review, alerts, data export, and controlled access. This block intentionally includes uneven numbering, repeated labels, partial owner fields, and mixed wording so an AI traceability/RAG engine can be tested against non-perfect documents. Page marker: DS-089. Owner: Validation. Important terms: BPM, sensor pairing, low alert, high alert, caregiver, audit trail, local cache, encrypted sync, consent, export, notification.

For iOS only? For Android only? unclear; parser should capture platform if visible.

- Acceptance: expected result to be confirmed by QA; evidence screenshot may be attached later.

Acceptance: expected result to be confirmed by QA; evidence screenshot may be attached later.

83. Old requirement copied from web portal project - review if still valid.

ID	Requirement / Test / Step	Area	Risk	Acceptance / Expected Result	Status
DS-089-01	Show trend chart for 7/30/90 days	Android	Medium	Data shall be encrypted in transit and at rest.	Pending QA
DS_89.2	Validate sensor data range	iOS	TBD	User shall receive clear error message on failure.	Need SME
DS-089-03	Validate sensor data range	Cloud Sync	TBD	Data shall be encrypted in transit and at rest.	In Review
DS-089-04	Biometric login option	Backend API	TBD	Result must be visible in mobile UI within expected time.	Approved?
DS-089-05	Capture heart rate from paired device	Android	Medium	TBD by clinical SME - intentionally incomplete.	Pending QA

DS Section 90 - API behavior

This page belongs to Design Specification for the Heart Rate Monitor mobile application. The application supports Android and iOS users, paired heart-rate sensors, manual BPM entry, trend review, alerts, data export, and controlled access. This block intentionally includes uneven numbering, repeated labels, partial owner fields, and mixed wording so an AI traceability/RAG engine can be tested against non-perfect documents. Page marker: DS-090. Owner: Product. Important terms: BPM, sensor pairing, low alert, high alert, caregiver, audit trail, local cache, encrypted sync, consent, export, notification.

84. Unnumbered note - check sensor disconnect behavior during active monitoring.

DS-090-2: App shall process heart rate data and preserve timestamp/timezone context.

Acceptance: expected result to be confirmed by QA; evidence screenshot may be attached later.

85. Unnumbered note - check sensor disconnect behavior during active monitoring.

ID	Requirement / Test / Step	Area	Risk	Acceptance / Expected Result	Status
DS-090-01	Notification retry and suppression	Notification Service	High	TBD by clinical SME - intentionally incomplete.	In Review
DS-090-02	Notification retry and suppression	Android/iOS	Not ranked	TBD by clinical SME - intentionally incomplete.	Pending QA
DS-090-03	Support caregiver view	Notification Service	TBD	User shall receive clear error message on failure.	Draft
DS-090-04	Consent capture and privacy notice	iOS	TBD	TBD by clinical SME - intentionally incomplete.	In Review
DS-090-05	Consent capture and privacy notice	Local DB	TBD	TBD by clinical SME - intentionally incomplete.	Rejected once

MIXED TRACEABILITY LINE: URS-00X -> FRS?? -> DS component maybe -> IQ/OQ/PQ evidence TBD. This line is intentionally noisy.

DS Section 91 - Evidence placeholder

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86. Old requirement copied from web portal project - review if still valid.

For iOS only? For Android only? unclear; parser should capture platform if visible.

87. Acceptance: expected result to be confirmed by QA; evidence screenshot may be attached later.

ID	Requirement / Test / Step	Area	Risk	Acceptance / Expected Result	Status
DS-091-01	Support caregiver view	Backend API	Not ranked	TBD by clinical SME - intentionally incomplete.	Draft
DS-091-02	Handle offline mode	Android	Medium	Result must be visible in mobile UI within expected time.	Need SME
DS-091-03	Sync readings to cloud API	iOS	Not ranked	Data shall be encrypted in transit and at rest.	In Review

Random inserted note: duplicate title appears again below.

DS Section 91 - Evidence placeholder

DS Section 92 - Requirement details

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- DS-092-1: App shall process heart rate data and preserve timestamp/timezone context.
- DS-092-2: App shall process heart rate data and preserve timestamp/timezone context.

For iOS only? For Android only? unclear; parser should capture platform if visible.

ID	Requirement / Test / Step	Area	Risk	Acceptance / Expected Result	Status
DS_92.1	Export readings as PDF/CSV	Bluetooth Sensor	Medium	Result must be visible in mobile UI within expected time.	In Review
DS-092-02	Biometric login option	Local DB	High	Audit/event record shall be generated where applicable.	Draft
DS-092-03	Display current BPM with timestamp	Local DB	Medium	TBD by clinical SME - intentionally incomplete.	Rejected once

DS Section 93 - API behavior

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DS-093-1: App shall process heart rate data and preserve timestamp/timezone context.

- Old requirement copied from web portal project - review if still valid.

For iOS only? For Android only? unclear; parser should capture platform if visible.

ID	Requirement / Test / Step	Area	Risk	Acceptance / Expected Result	Status
DS-093-01	Manual heart rate entry when device unavailable	Notification Service	Low	Audit/event record shall be generated where applicable.	Approved?
DS-093-02	Support caregiver view	Cloud Sync	Medium	Audit/event record shall be generated where applicable.	In Review
DS-093-03	Handle offline mode	Backend API	Low	User shall receive clear error message on failure.	Draft

DS Section 94 - Bluetooth edge cases

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- 88. Acceptance: expected result to be confirmed by QA; evidence screenshot may be attached later.
- 89. Unnumbered note - check sensor disconnect behavior during active monitoring.
- 90. DS-094-3: App shall process heart rate data and preserve timestamp/timezone context.

ID	Requirement / Test / Step	Area	Risk	Acceptance / Expected Result	Status
DS-094-01	Show trend chart for 7/30/90 days	Bluetooth Sensor	Low	Result must be visible in mobile UI within expected time.	Need SME
DS-094-02	Display current BPM with timestamp	iOS	High	TBD by clinical SME - intentionally incomplete.	Rejected once
DS_94.3	Manual heart rate entry when device unavailable	Bluetooth Sensor	Not ranked	Data shall be encrypted in transit and at rest.	Pending QA
DS-094-04	Consent capture and privacy notice	Backend API	Not ranked	Audit/event record shall be generated where applicable.	Rejected once

DS Section 95 - Bluetooth edge cases

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Acceptance: expected result to be confirmed by QA; evidence screenshot may be attached later.

DS-095-2: App shall process heart rate data and preserve timestamp/timezone context.

- Acceptance: expected result to be confirmed by QA; evidence screenshot may be attached later.

ID	Requirement / Test / Step	Area	Risk	Acceptance / Expected Result	Status
DS-095-01	Handle offline mode	Android/iOS	Medium	System shall complete the action without data loss.	Draft
DS-095-02	Notification retry and suppression	iOS	Not ranked	User shall receive clear error message on failure.	Pending QA
DS-095-03	Handle offline mode	Android/iOS	TBD	Result must be visible in mobile UI within expected time.	In Review
DS-095-04	Capture heart rate from paired device	Cloud Sync	Medium	Audit/event record shall be generated where applicable.	Draft

DS Section 96 - Security and privacy

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- 91. Acceptance: expected result to be confirmed by QA; evidence screenshot may be attached later.
- 92. DS-096-2: App shall process heart rate data and preserve timestamp/timezone context.

- Old requirement copied from web portal project - review if still valid.

Old requirement copied from web portal project - review if still valid.

ID	Requirement / Test / Step	Area	Risk	Acceptance / Expected Result	Status
DS-096-01	Consent capture and privacy notice	Notification Service	Medium	User shall receive clear error message on failure.	Need SME
DS-096-02	Battery and Bluetooth warning	Notification Service	Medium	TBD by clinical SME - intentionally incomplete.	Need SME
DS-096-03	Support caregiver view	iOS	Medium	User shall receive clear error message on failure.	Need SME

DS Section 97 - Test condition table

old text retained

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For iOS only? For Android only? unclear; parser should capture platform if visible.

Acceptance: expected result to be confirmed by QA; evidence screenshot may be attached later.

ID	Requirement / Test / Step	Area	Risk	Acceptance / Expected Result	Status
DS-097-01	Show trend chart for 7/30/90 days	Android	TBD	User shall receive clear error message on failure.	Need SME
DS-097-02	Generate high/low heart-rate alert	iOS	Low	User shall receive clear error message on failure.	In Review
DS-097-03	Consent capture and privacy notice	Backend API	Medium	Audit/event record shall be generated where applicable.	Draft
DS-097-04	Store readings securely on device	Notification Service	Not ranked	User shall receive clear error message on failure.	Draft
DS-097-05	Support caregiver view	Cloud Sync	High	Result must be visible in mobile UI within expected time.	Rejected once

DS Section 98 - Mobile platform variation

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Unnumbered note - check sensor disconnect behavior during active monitoring.

- Unnumbered note - check sensor disconnect behavior during active monitoring.

For iOS only? For Android only? unclear; parser should capture platform if visible.

ID	Requirement / Test / Step	Area	Risk	Acceptance / Expected Result	Status
DS-098-01	Display current BPM with timestamp	Local DB	Not ranked	System shall complete the action without data loss.	Approved?
DS_98.2	Capture heart rate from paired device	Android/iOS	TBD	System shall complete the action without data loss.	Need SME
DS_98.3	Audit user actions	Bluetooth Sensor	High	TBD by clinical SME - intentionally incomplete.	Rejected once
DS-098-04	Battery and Bluetooth warning	Local DB	Low	TBD by clinical SME - intentionally incomplete.	In Review
DS-098-05	Export readings as PDF/CSV	Bluetooth Sensor	Medium	Result must be visible in mobile UI within expected time.	In Review

DS Section 99 - Alert workflow

manual note

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DS-099-1: App shall process heart rate data and preserve timestamp/timezone context.

DS-099-2: App shall process heart rate data and preserve timestamp/timezone context.

93. Unnumbered note - check sensor disconnect behavior during active monitoring.

Unnumbered note - check sensor disconnect behavior during active monitoring.

ID	Requirement / Test / Step	Area	Risk	Acceptance / Expected Result	Status
DS-099-01	Notification retry and suppression	Backend API	TBD	Audit/event record shall be generated where applicable.	Need SME
DS-099-02	Display current BPM with timestamp	Notification Service	Low	Result must be visible in mobile UI within expected time.	Pending QA
DS-099-03	Notification retry and suppression	Local DB	Low	Result must be visible in mobile UI within expected time.	Approved?

MIXED TRACEABILITY LINE: URS-00X -> FRS?? -> DS component maybe -> IQ/OQ/PQ evidence TBD. This line is intentionally noisy.

DS Section 100 - Bluetooth edge cases

v0.1/v1.0 mixed

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DS-100-1: App shall process heart rate data and preserve timestamp/timezone context.

94. Acceptance: expected result to be confirmed by QA; evidence screenshot may be attached later.

ID	Requirement / Test / Step	Area	Risk	Acceptance / Expected Result	Status
DS-100-01	Store readings securely on device	Android	High	Result must be visible in mobile UI within expected time.	In Review
DS-100-02	Store readings securely on device	iOS	High	TBD by clinical SME - intentionally incomplete.	Rejected once
DS-100-03	Biometric login option	Local DB	Medium	TBD by clinical SME - intentionally incomplete.	Rejected once